# Lab Assignment

## Installing Kali Linux on VMware Workstation

A Step-by-Step Guide with Screenshots

## Objective

The objective of this lab is to install Kali Linux as a virtual machine inside VMware Workstation (or VMware Player). By the end of this lab, students will be able to download Kali Linux ISO, create a new virtual machine in VMware, configure hardware settings, install the operating system, and log in successfully.

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## System Requirements

| Component | Requirement |
|---|---|
| **Processor** | 64-bit Intel Core i3/i5/i7 or AMD equivalent |
| **RAM** | 4 GB minimum (8 GB recommended for smooth performance) |
| **Storage** | 30 GB free disk space minimum |
| **Operating System** | Windows 10/11 (64-bit) or Linux host |
| **VMware Version** | VMware Workstation Pro/Player 16 or later |
| **Kali Linux ISO** | Latest version from kali.org (e.g., kali-linux-2024.x-amd64.iso) |
| **Internet** | Required to download ISO and VMware tools |

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## Steps

### Step 1: Download Kali Linux ISO File

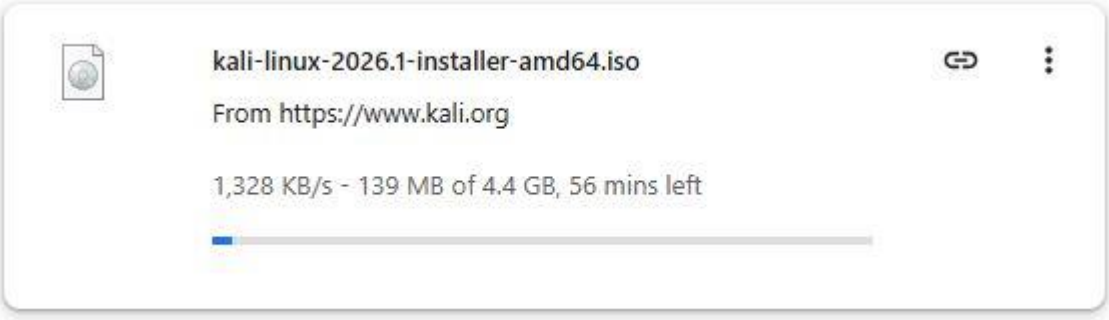
- Open your browser and go to the official Kali Linux website:

https://www.kali.org/get-kali/#kali-installer-images

- Under the Installer Images section, click on the 64-bit ISO download link.
- Wait for the download to complete. The file size is typically around 3–4 GB.
- Optionally verify the SHA256 checksum to ensure the file is not corrupted.

Installer Images
Direct access to hardware
Customized Kali kernel
No overhead
Single or multiple boot Kali, giving you complete control over the hardware access (perfect for in-built Wi-Fi and GPU), enabling the best performance.
Recommended

Kali Linux 2026.1 Changelog
x86_64
Apple Silicon (ARM64)
Recommended
Installer
Complete offline installation with customization
4.4G
torrent
sum

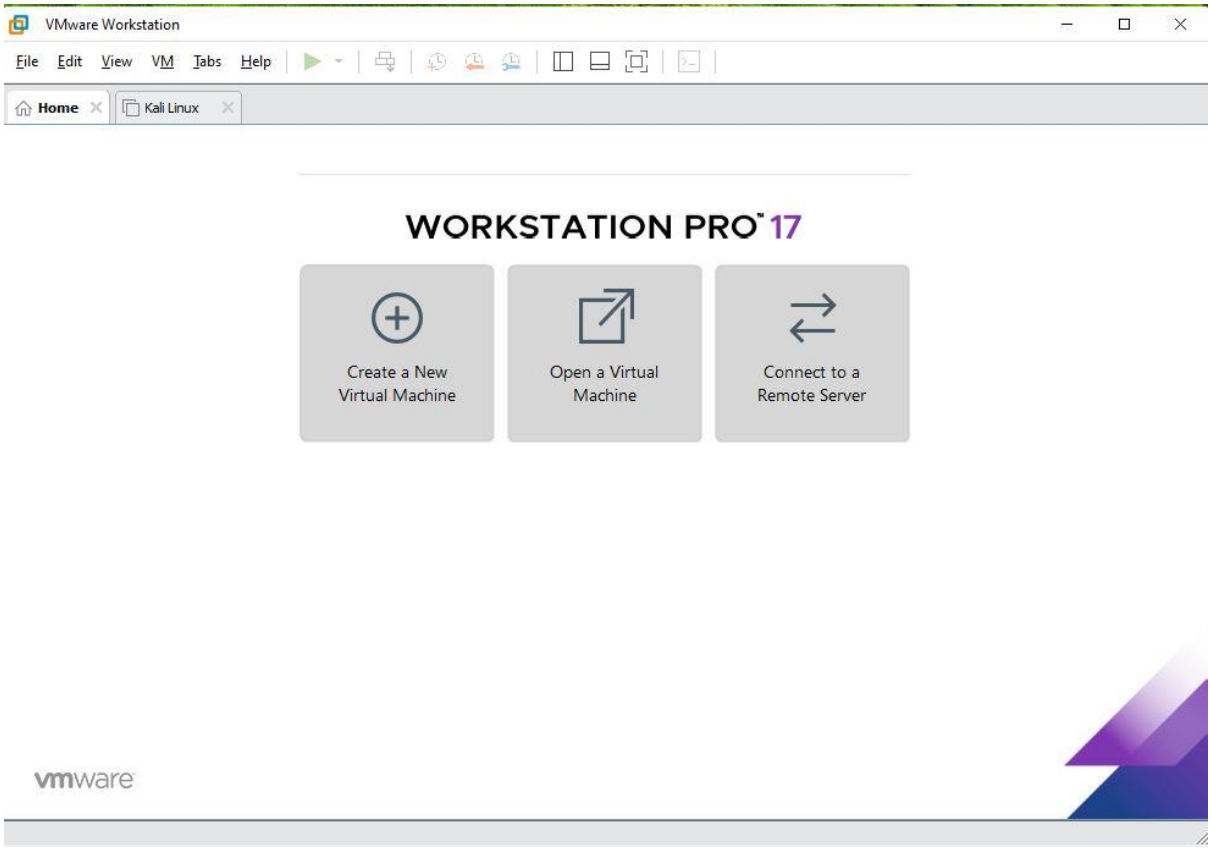


## Step 2: Download and Install VMware Workstation

- Go to the VMware official website:

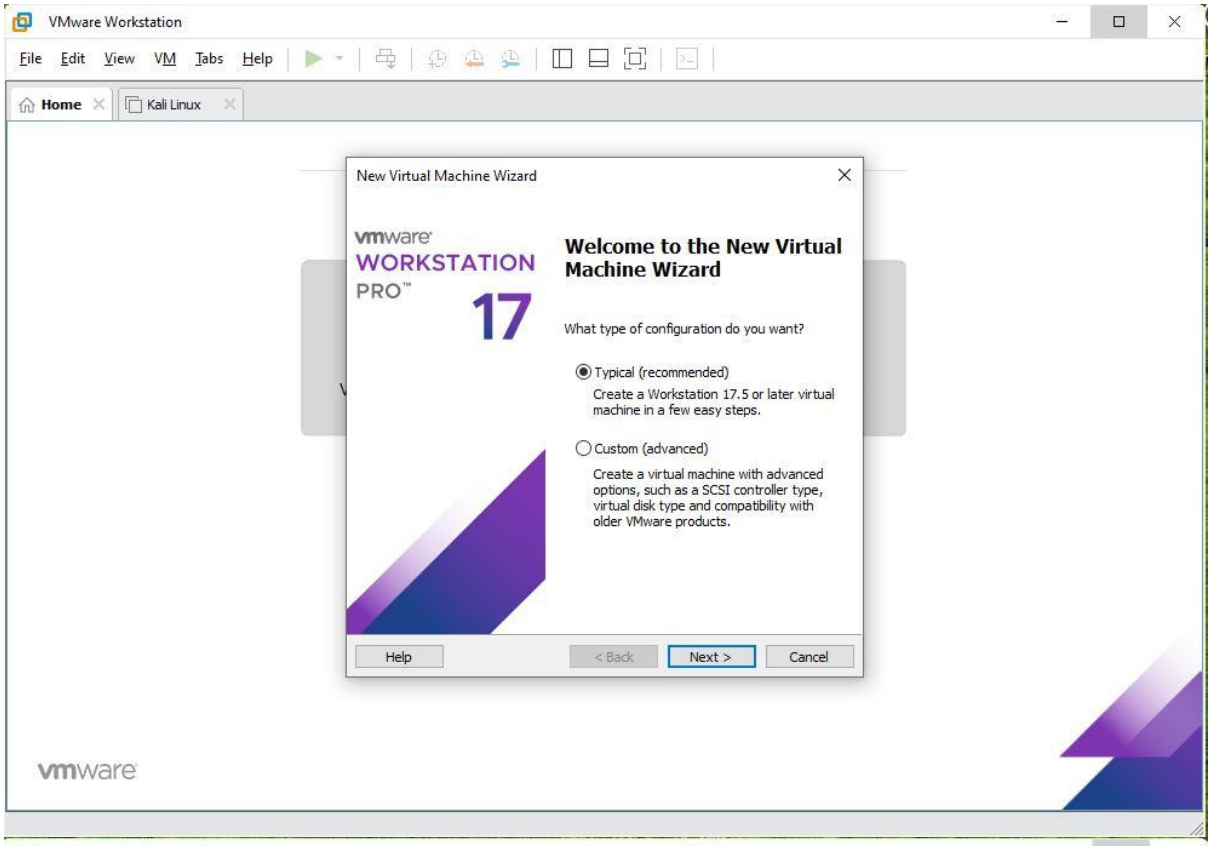
https://www.vmware.com/products/workstation-player/workstation-player-evaluation.html

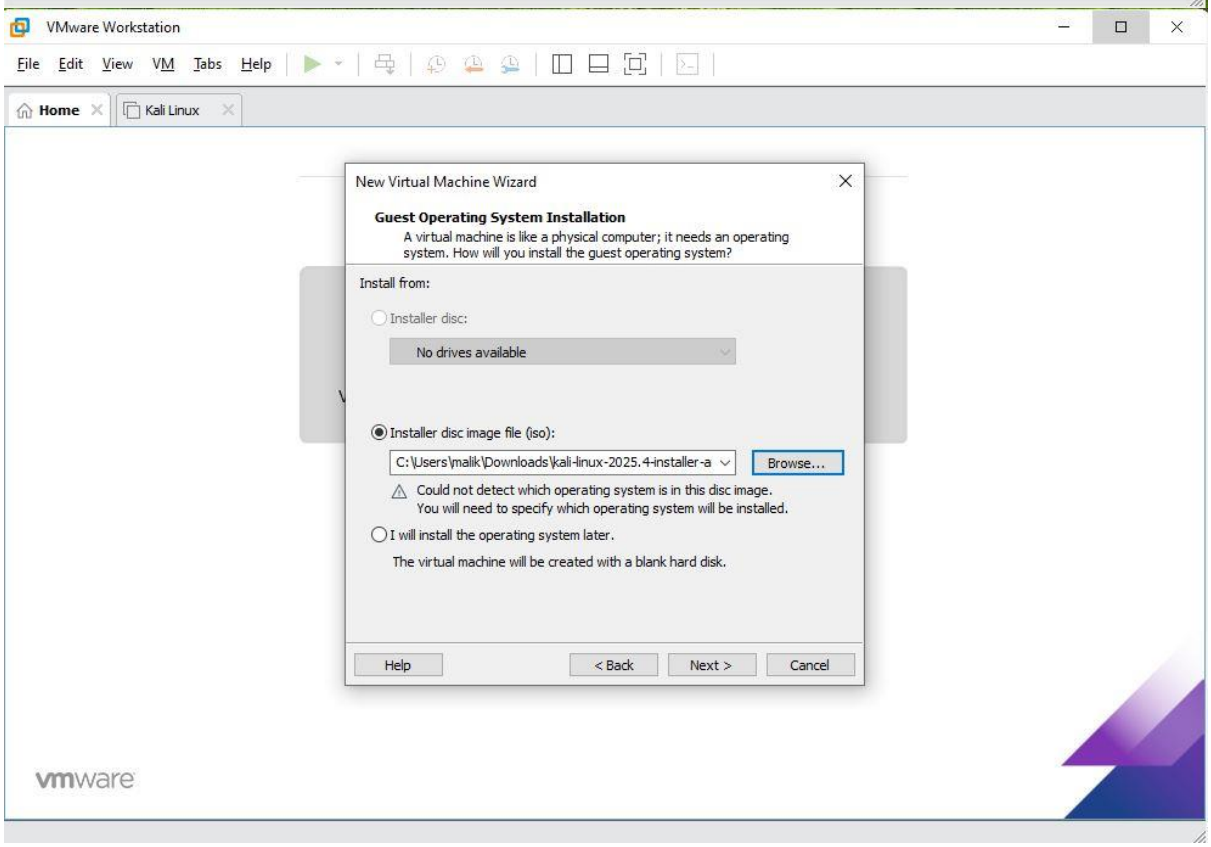
- Download VMware Workstation Player (free for personal use) or VMware Workstation Pro (paid/trial).
- Run the downloaded installer (.exe on Windows).
- Follow the on-screen setup wizard: Accept license agreement → Choose installation path → Click Install.
- Restart your computer after installation if prompted.




## Step 3: Launch VMware and Create a New Virtual Machine

- Open VMware Workstation Player or Pro from the desktop shortcut or Start Menu.
- On the home screen, click on 'Create a New Virtual Machine'.
- The New Virtual Machine Wizard will open.
- Select 'Installer disc image file (iso)' and click Browse to locate the Kali Linux ISO you downloaded.
- VMware may detect it as 'Linux' automatically — click Next to continue.

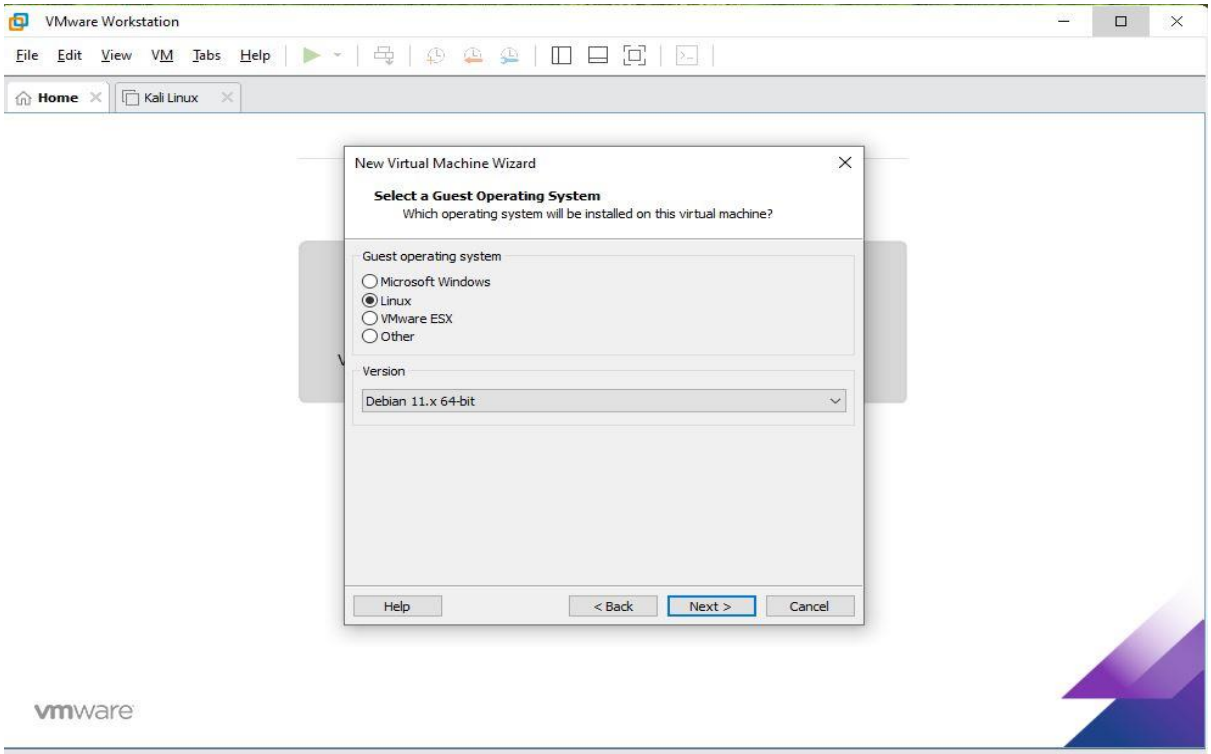


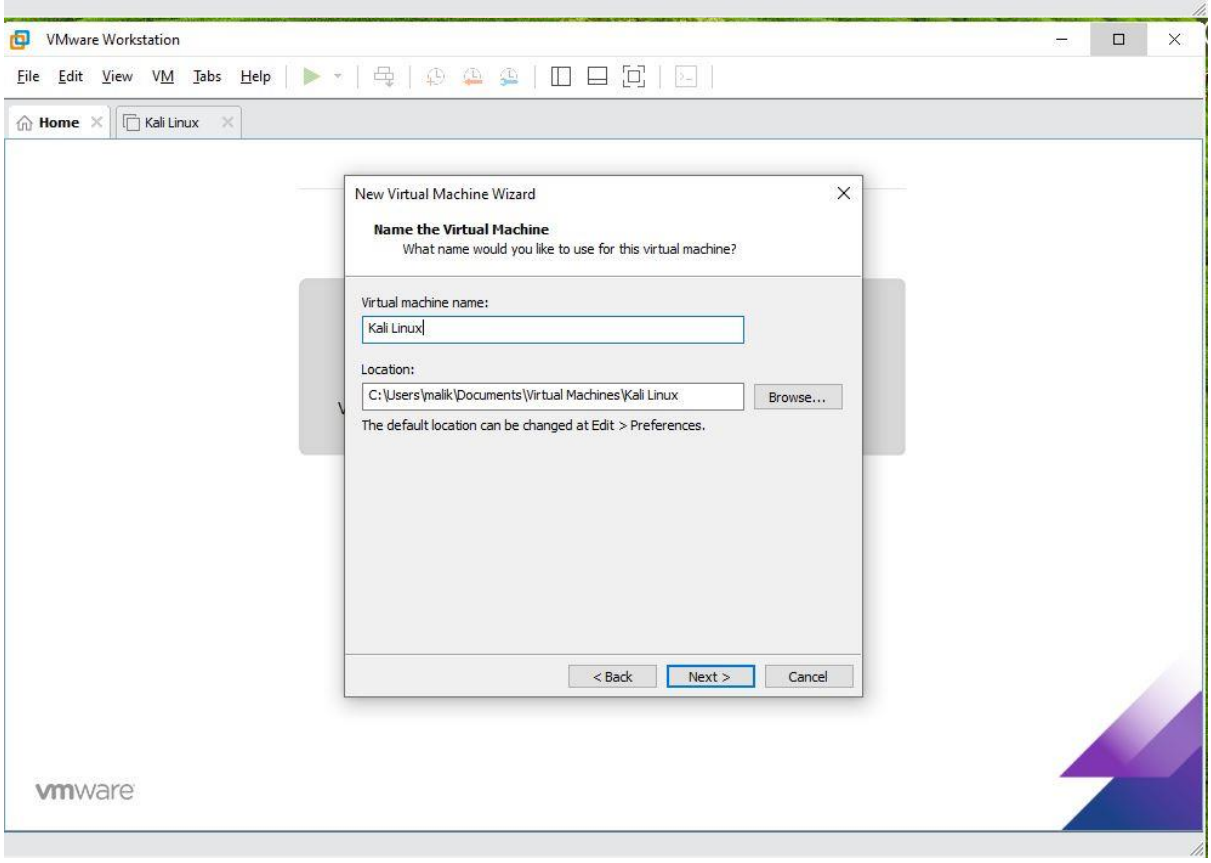



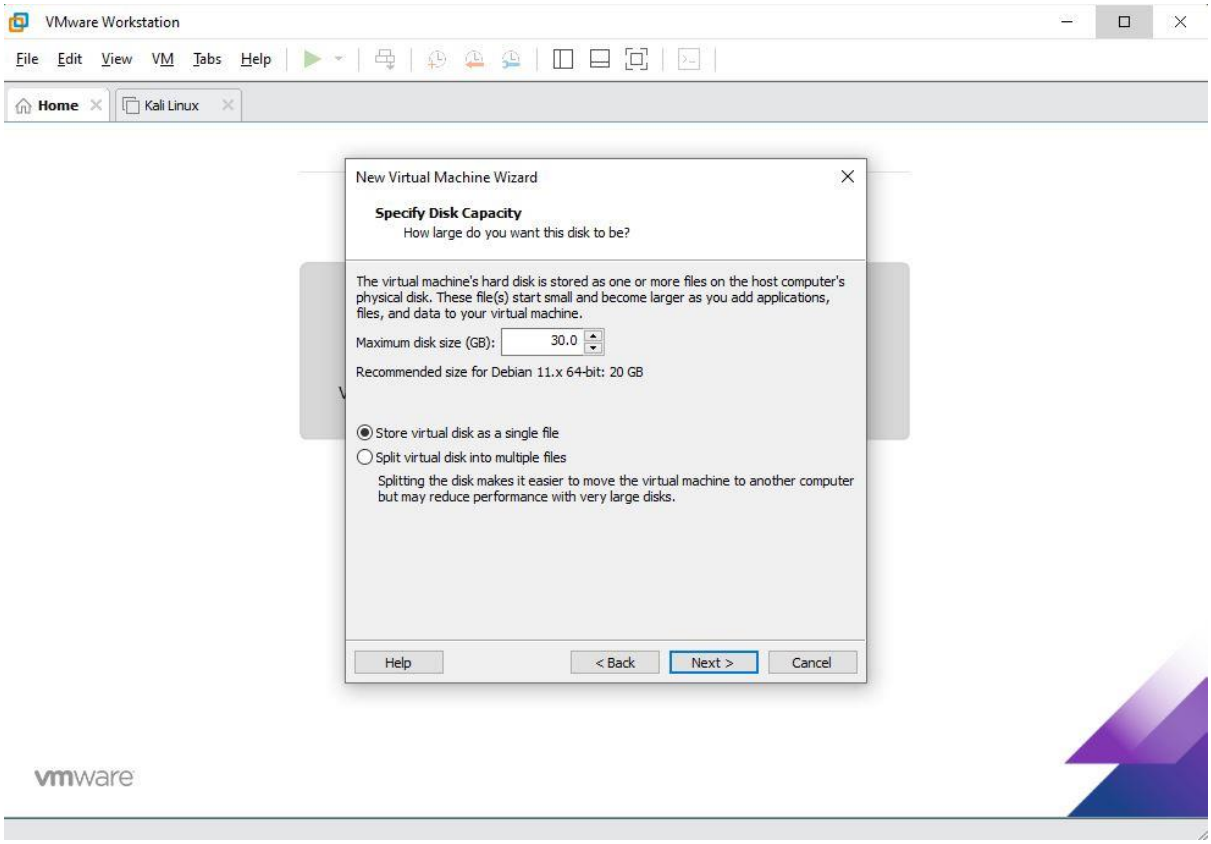

## Step 4: Configure Virtual Machine Settings

- Guest Operating System: Select Linux, then from the Version dropdown choose 'Debian 11.x 64-bit' (Kali is based on Debian).

- Virtual Machine Name: Type a name (e.g., Kali-Linux-2024) and choose a folder to save the VM.
- Processor Configuration: Set Number of Processors to 1 and Number of cores per processor to 2 (or more if your CPU supports it).
- Memory (RAM): Allocate at least 4096 MB (4 GB). 8 GB is recommended for best performance.
- Network Type: Select 'Use Network Address Translation (NAT)' so the VM can share the host internet.
- Disk: Create a new virtual disk with at least 30 GB. Choose 'Store virtual disk as a single file' for simplicity.

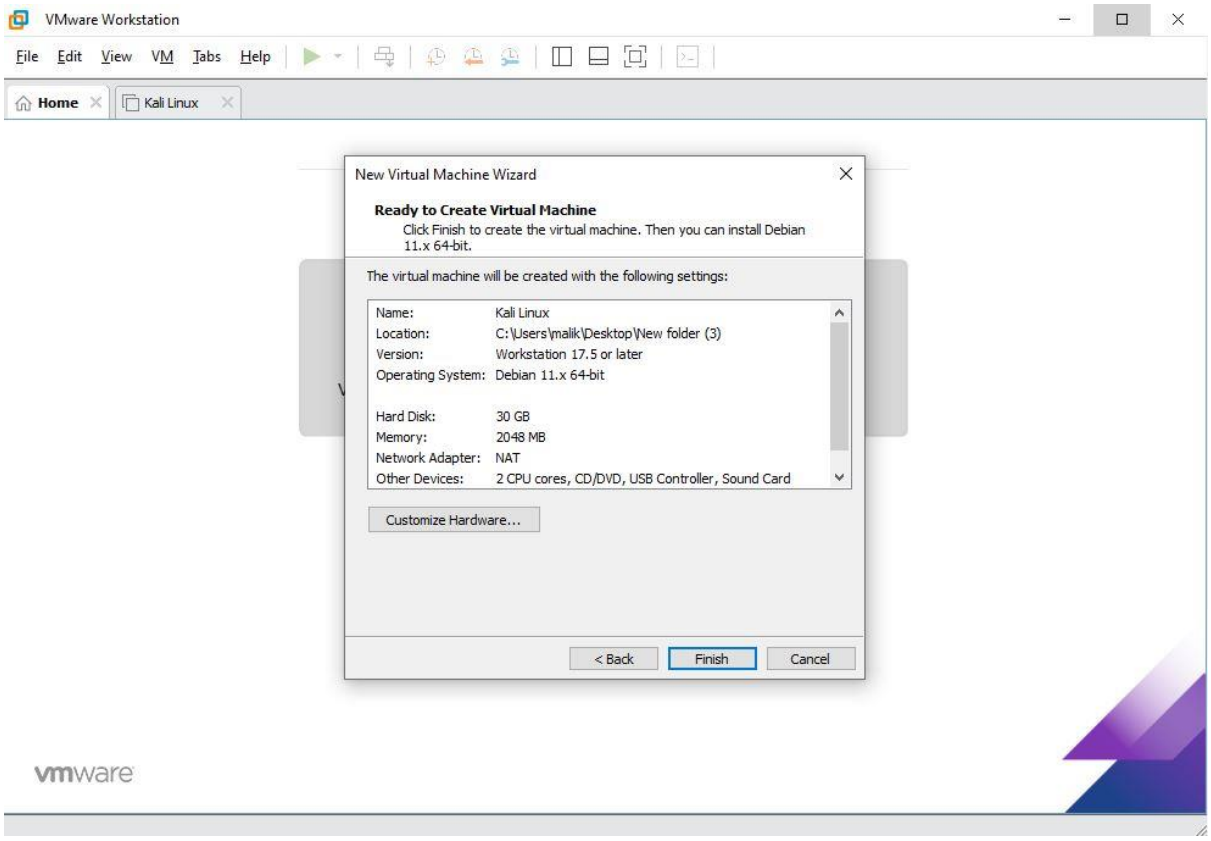

VMware Workstation
File Edit View VM Tabs Help
Home
Kali Linux
New Virtual Machine Wizard
Select a Guest Operating System
Which operating system will be installed on this virtual machine?
Guest operating system
Microsoft Windows
Linux
VMware ESX
Other
Version
Debian 11.x 64-bit
Help
< Back
Next >
Cancel
vmware


VMware Workstation
File Edit View VM Tabs Help
Home
Kali Linux
New Virtual Machine Wizard
Name the Virtual Machine
What name would you like to use for this virtual machine?
Virtual machine name:
Kali Linux
Location:
C:\Users\malik\Documents\Virtual Machines\Kali Linux
Browse...
The default location can be changed at Edit > Preferences.
< Back
Next >
Cancel
vmware




## Step 5: Review Summary and Finish

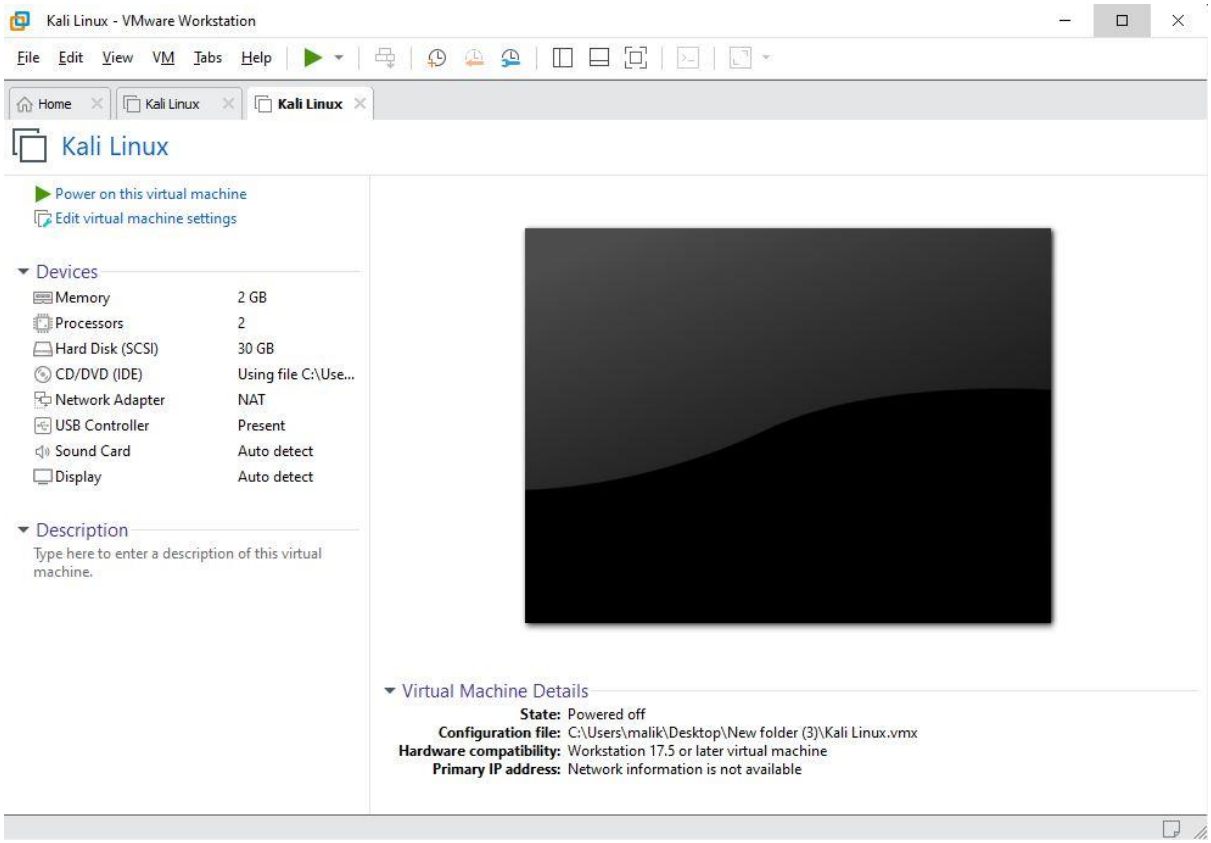
- Review the summary page showing all VM settings.
- Verify: ISO path, OS type, RAM, and Disk size are correct.
- Click 'Finish' to create the virtual machine.
- The new VM will appear in the VMware library on the left sidebar.




## Step 6: Power On the Virtual Machine

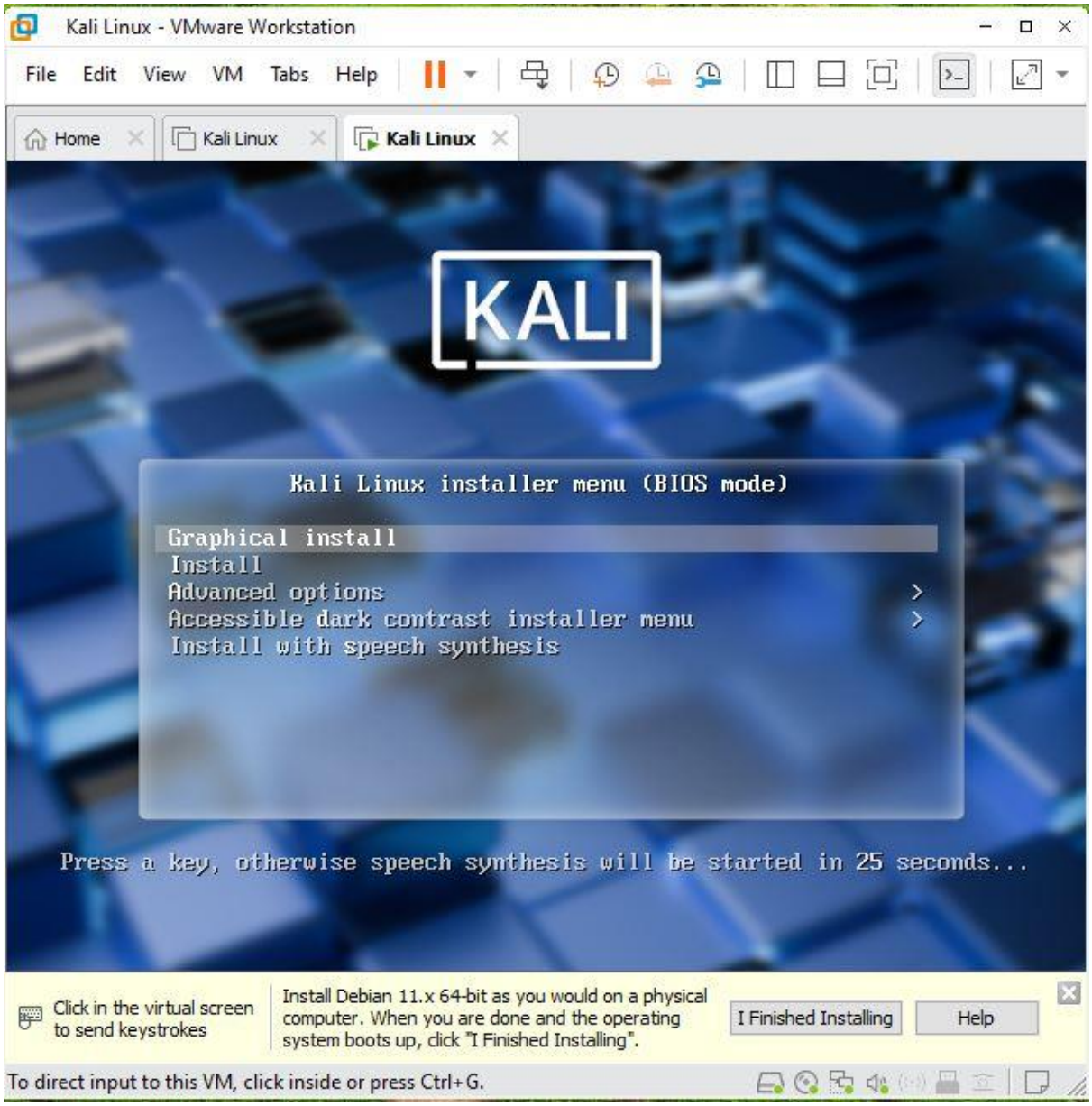
- Select your newly created Kali Linux VM from the list.
- Click the green 'Power On' button (or press Ctrl+B).
- The VM will start booting from the ISO image.
- A splash screen with the Kali Linux logo will appear.

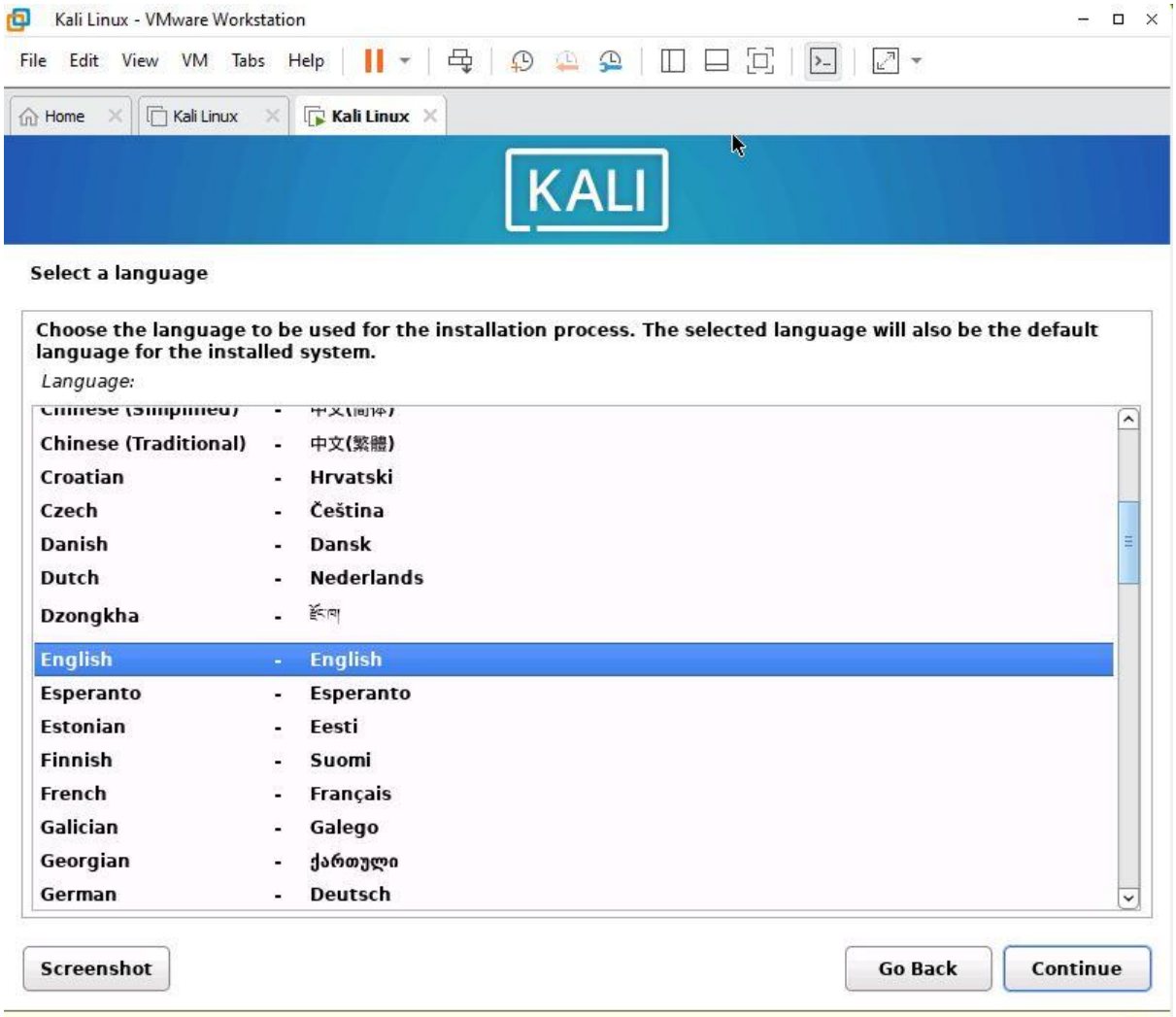
**Note:** If VMware asks to install VMware Tools, click 'Remind Me Later' — you can do this after installation.

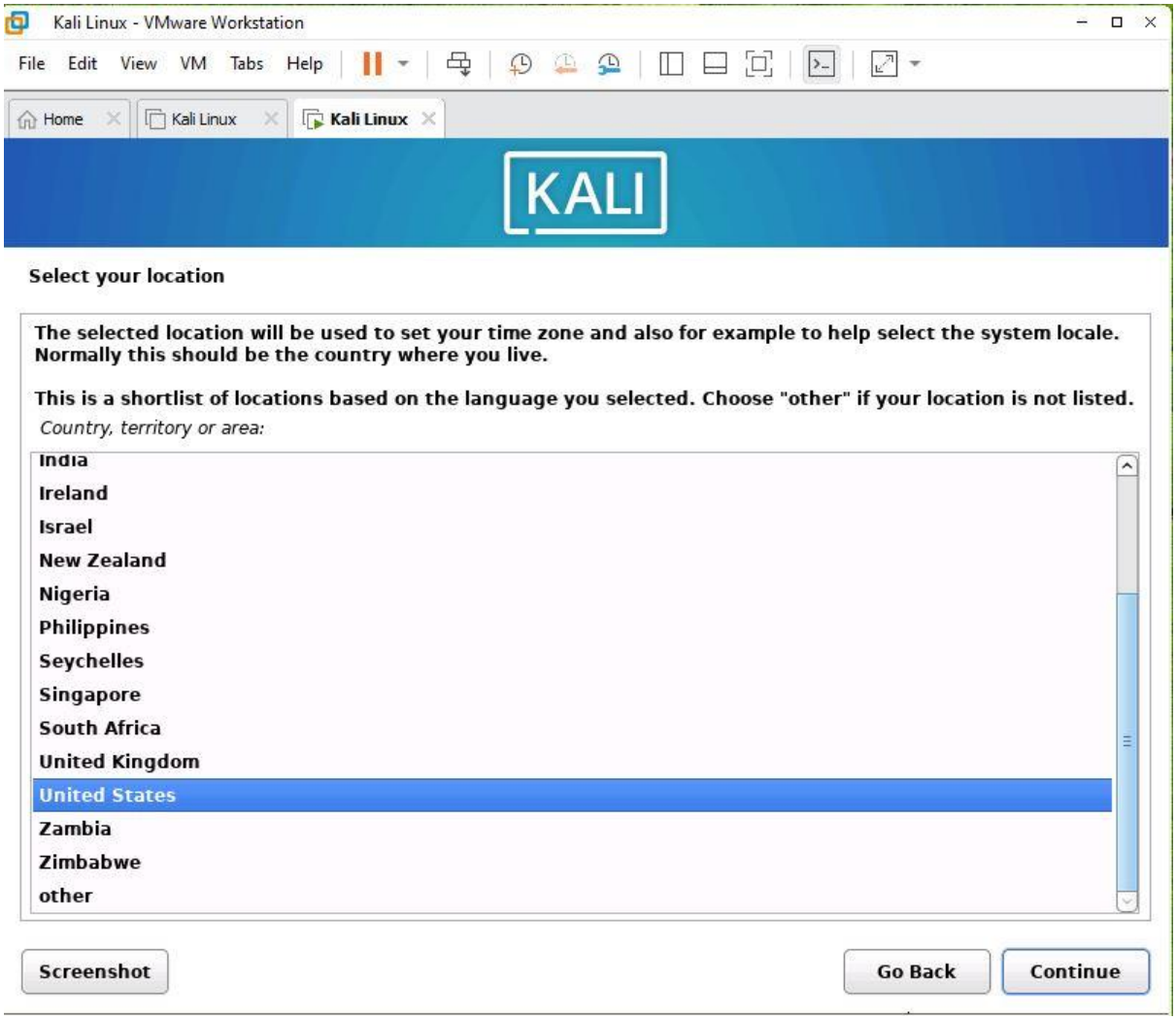


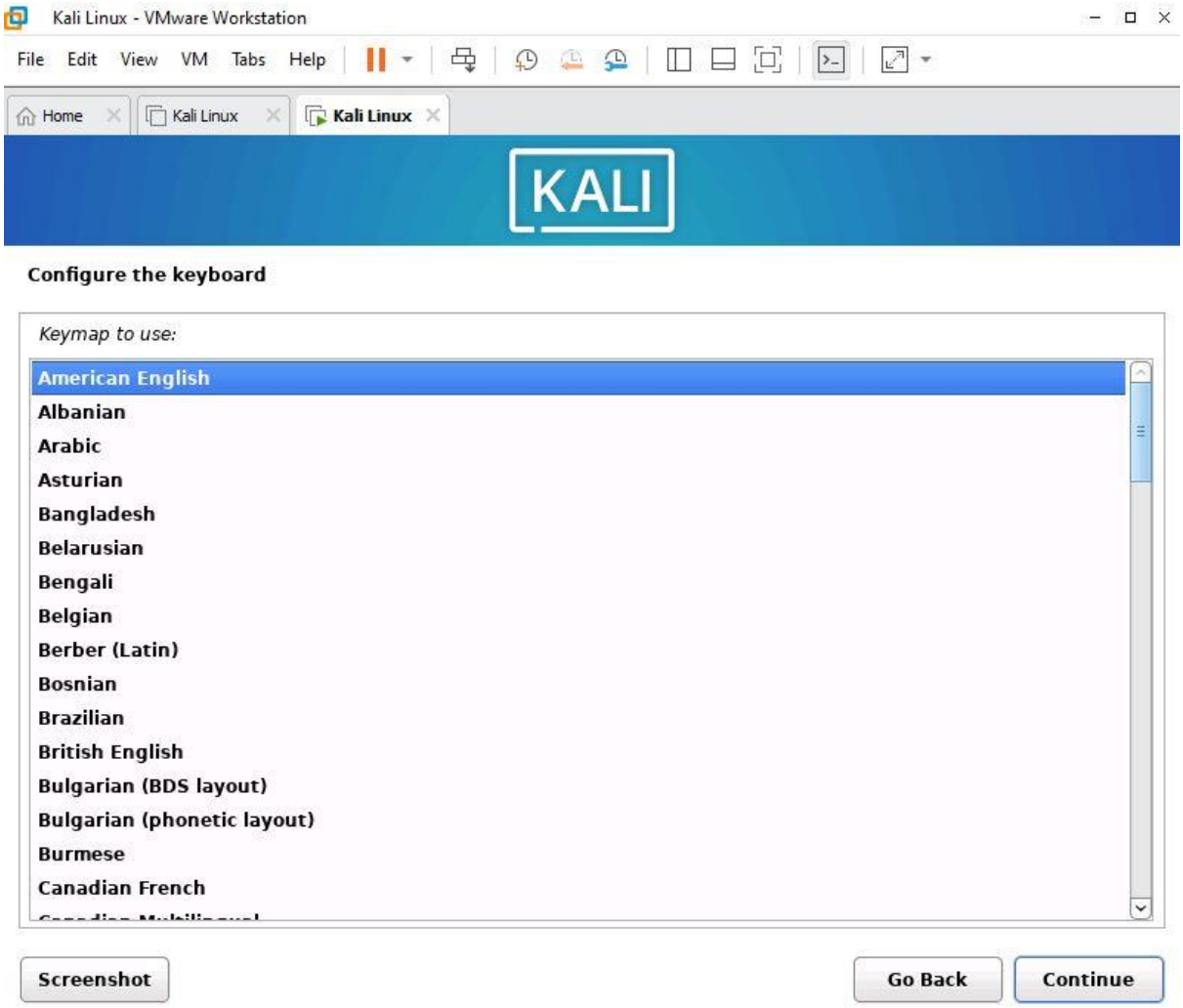

## Step 7: Start the Kali Linux Graphical Installation

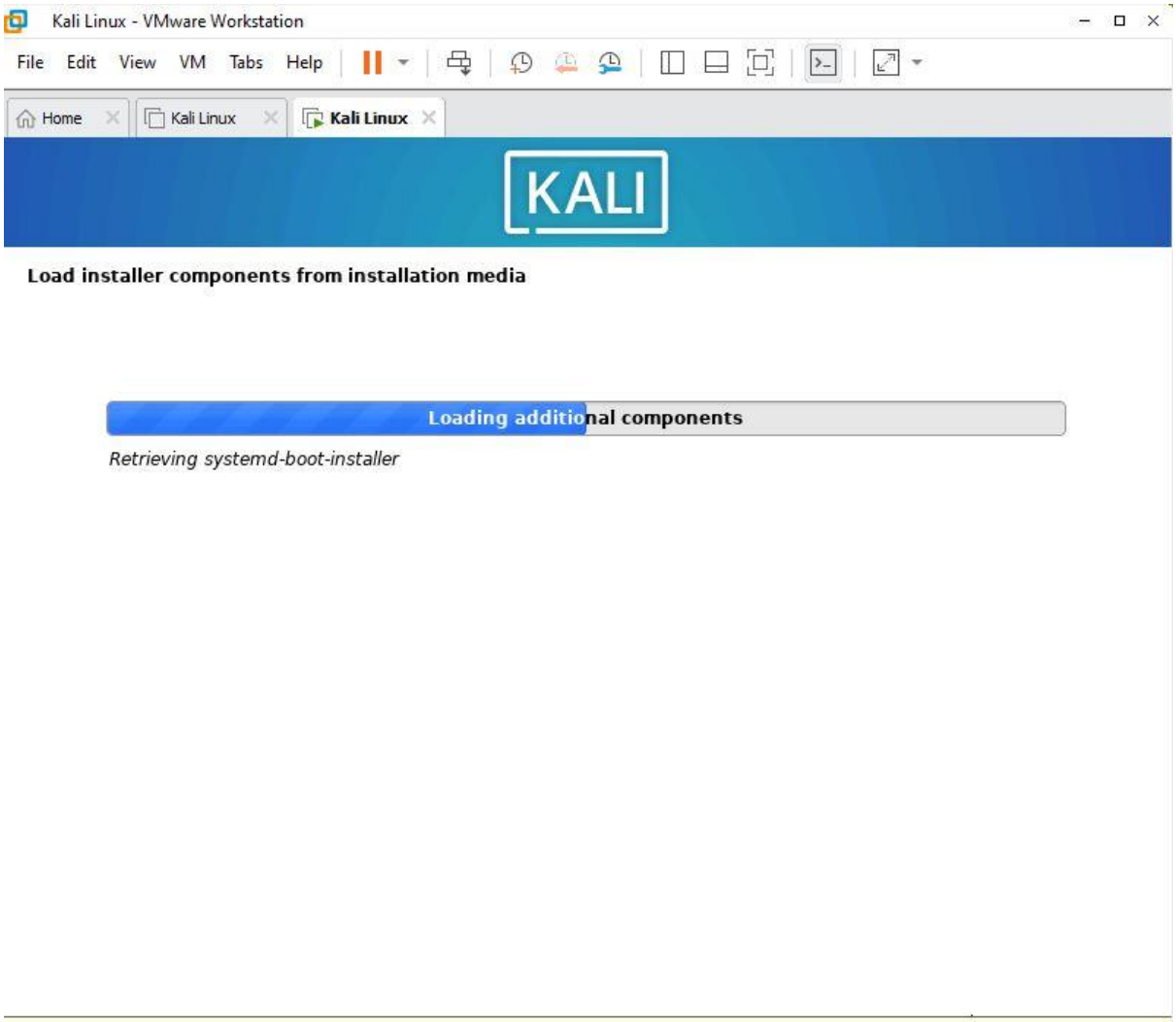
- From the Kali boot menu, use the arrow keys to select 'Graphical Install' and press Enter.
- Language: Select your preferred language (e.g., English) and click Continue.
- Location: Choose your country or region and click Continue.
- Keyboard: Select your keyboard layout (e.g., American English) and click Continue.


Kali Linux - VMware Workstation
File Edit View VM Tabs Help
Home
Kali Linux
Kali Linux
KALI
Kali Linux installer menu (BIOS mode)
Graphical install
Install
Advanced options
Accessible dark contrast installer menu
Install with speech synthesis
Press a key, otherwise speech synthesis will be started in 25 seconds...
Click in the virtual screen to send keystrokes
Install Debian 11.x 64-bit as you would on a physical computer. When you are done and the operating system boots up, click "I Finished Installing".
I Finished Installing
Help
To direct input to this VM, click inside or press Ctrl+G.


Kali Linux - VMware Workstation
File Edit View VM Tabs Help
Home
Kali Linux
Kali Linux
KALI
Select a language
Choose the language to be used for the installation process. The selected language will also be the default language for the installed system.
Language:
Chinese (Traditional) - 中文(繁體)
Croatian - Hrvatski
Czech - Čeština
Danish - Dansk
Dutch - Nederlands
Dzongkha - ཇོང་ཁ
English - English
Esperanto - Esperanto
Estonian - Eesti
Finnish - Suomi
French - Français
Galician - Galego
Georgian - ქართული
German - Deutsch
Screenshot
Go Back
Continue

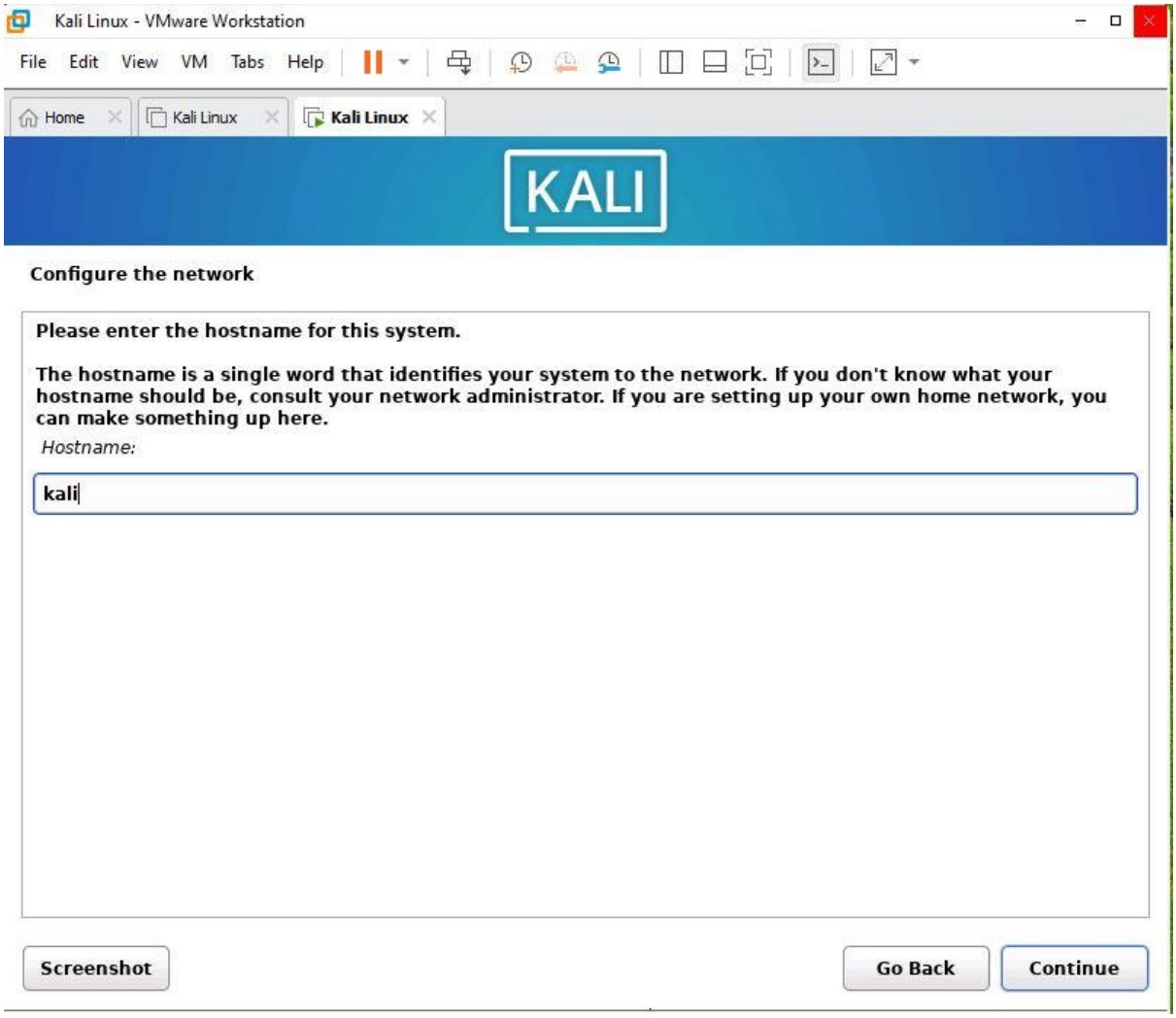

Kali Linux - VMware Workstation
File Edit View VM Tabs Help
Home
Kali Linux
Kali Linux
KALI
Select your location
The selected location will be used to set your time zone and also for example to help select the system locale. Normally this should be the country where you live.
This is a shortlist of locations based on the language you selected. Choose "other" if your location is not listed.
Country, territory or area:
India
Ireland
Israel
New Zealand
Nigeria
Philippines
Seychelles
Singapore
South Africa
United Kingdom
United States
Zambia
Zimbabwe
other
Screenshot
Go Back
Continue


Kali Linux - VMware Workstation
File Edit View VM Tabs Help
Home
Kali Linux
Kali Linux
KALI
Configure the keyboard
Keymap to use:
American English
Albanian
Arabic
Asturian
Bangladesh
Belarusian
Bengali
Belgian
Berber (Latin)
Bosnian
Brazilian
British English
Bulgarian (BDS layout)
Bulgarian (phonetic layout)
Burmese
Canadian French
Screenshot
Go Back
Continue




## Step 8: Configure Hostname and Network

- Hostname: Enter a hostname for your machine (e.g., kali) and click Continue.
- Domain Name: You may leave this blank or type 'local' and click Continue.
- The installer will automatically configure the network via DHCP (using VMware NAT).


Kali Linux - VMware Workstation
File Edit View VM Tabs Help
Home
Kali Linux
Kali Linux
KALI
Configure the network
Please enter the hostname for this system.
The hostname is a single word that identifies your system to the network. If you don't know what your hostname should be, consult your network administrator. If you are setting up your own home network, you can make something up here.
Hostname:
kali
Screenshot
Go Back
Continue

Kali Linux - VMware Workstation

File Edit View VM Tabs Help

Home | Kali Linux | Kali Linux

KALI

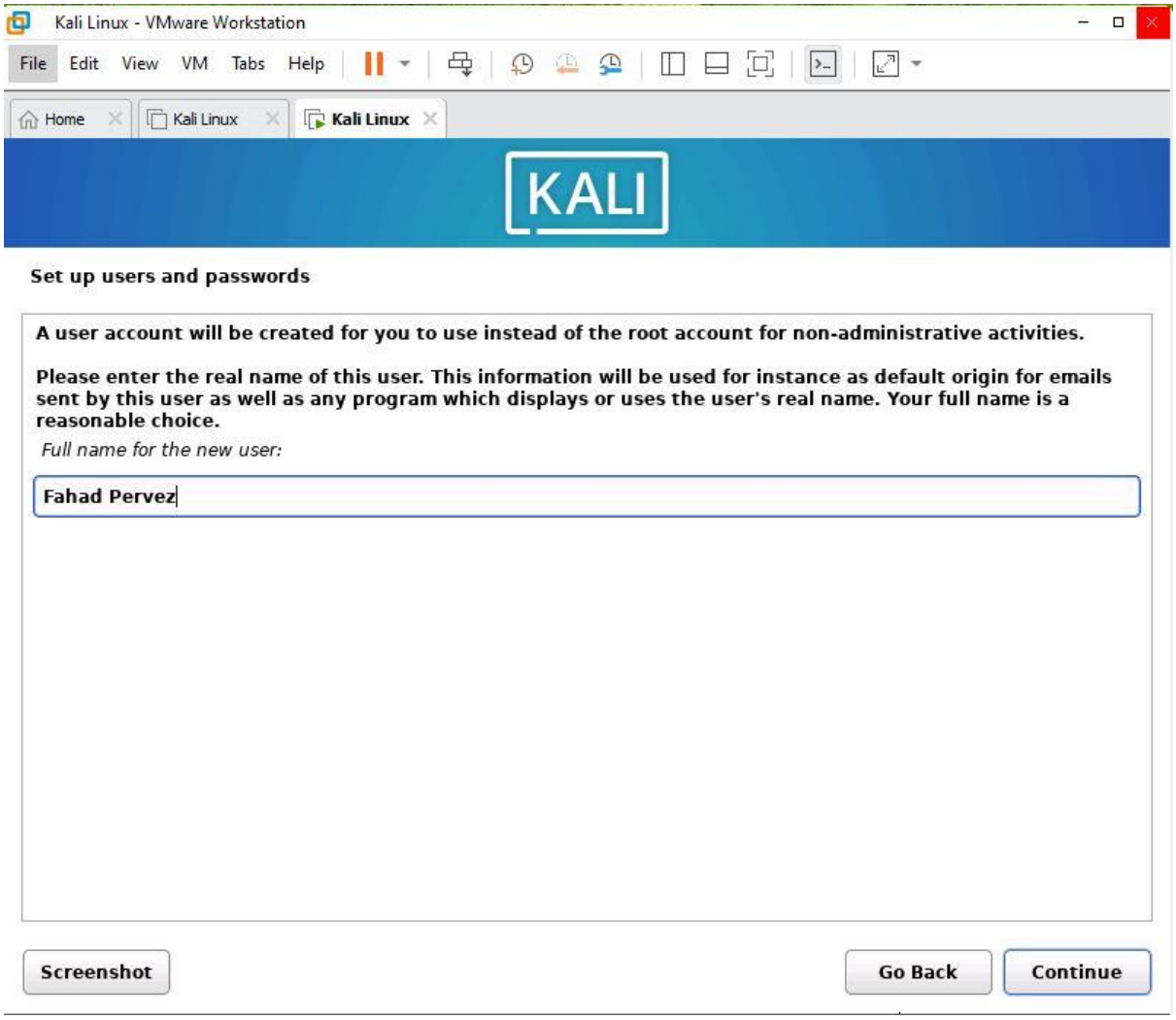
**Configure the network**

**The domain name is the part of your Internet address to the right of your host name. It is often something that ends in .com, .net, .edu, or .org. If you are setting up a home network, you can make something up, but make sure you use the same domain name on all your computers.**

*Domain name:*

localdomain

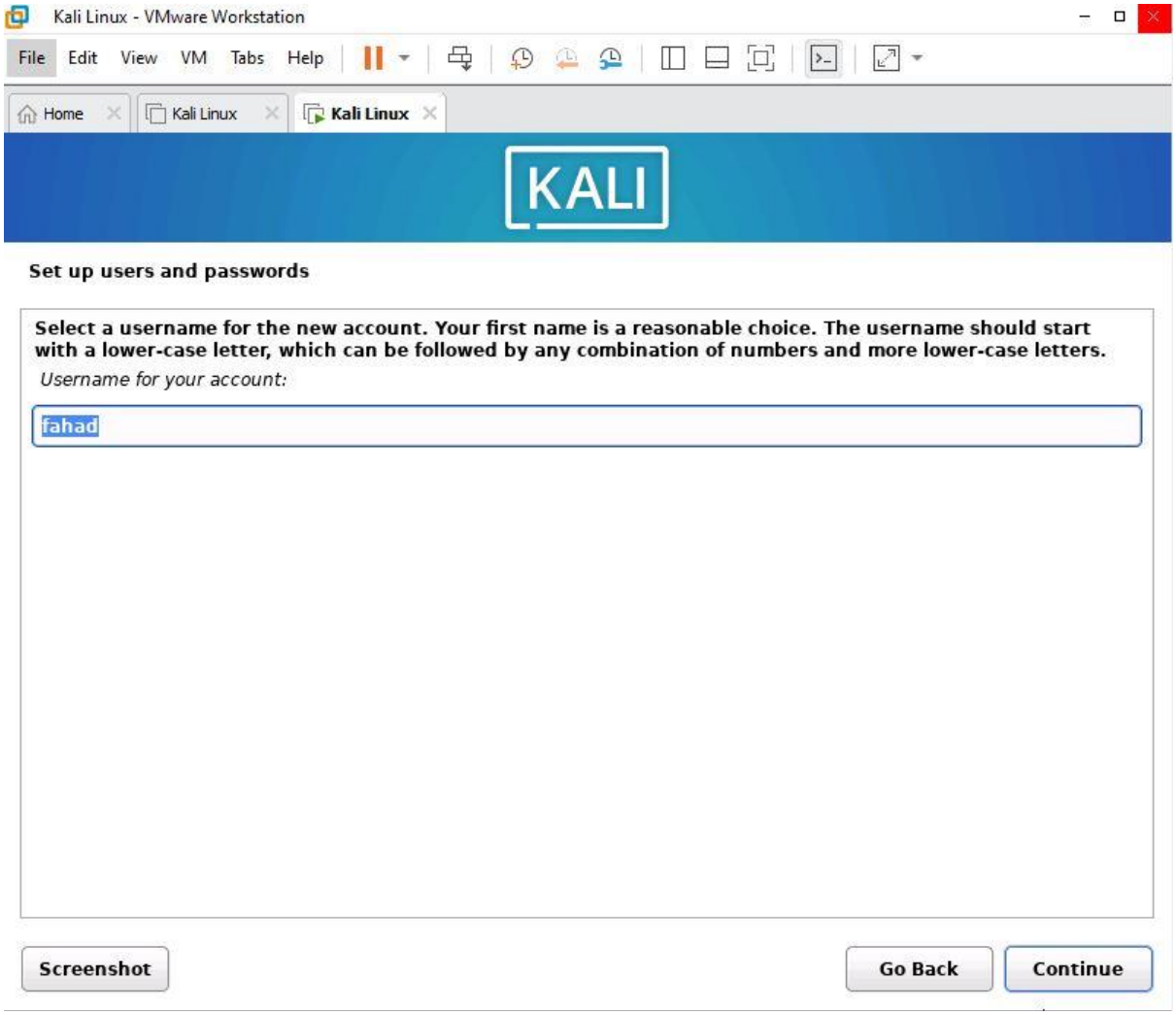
Screenshot | Go Back | Continue

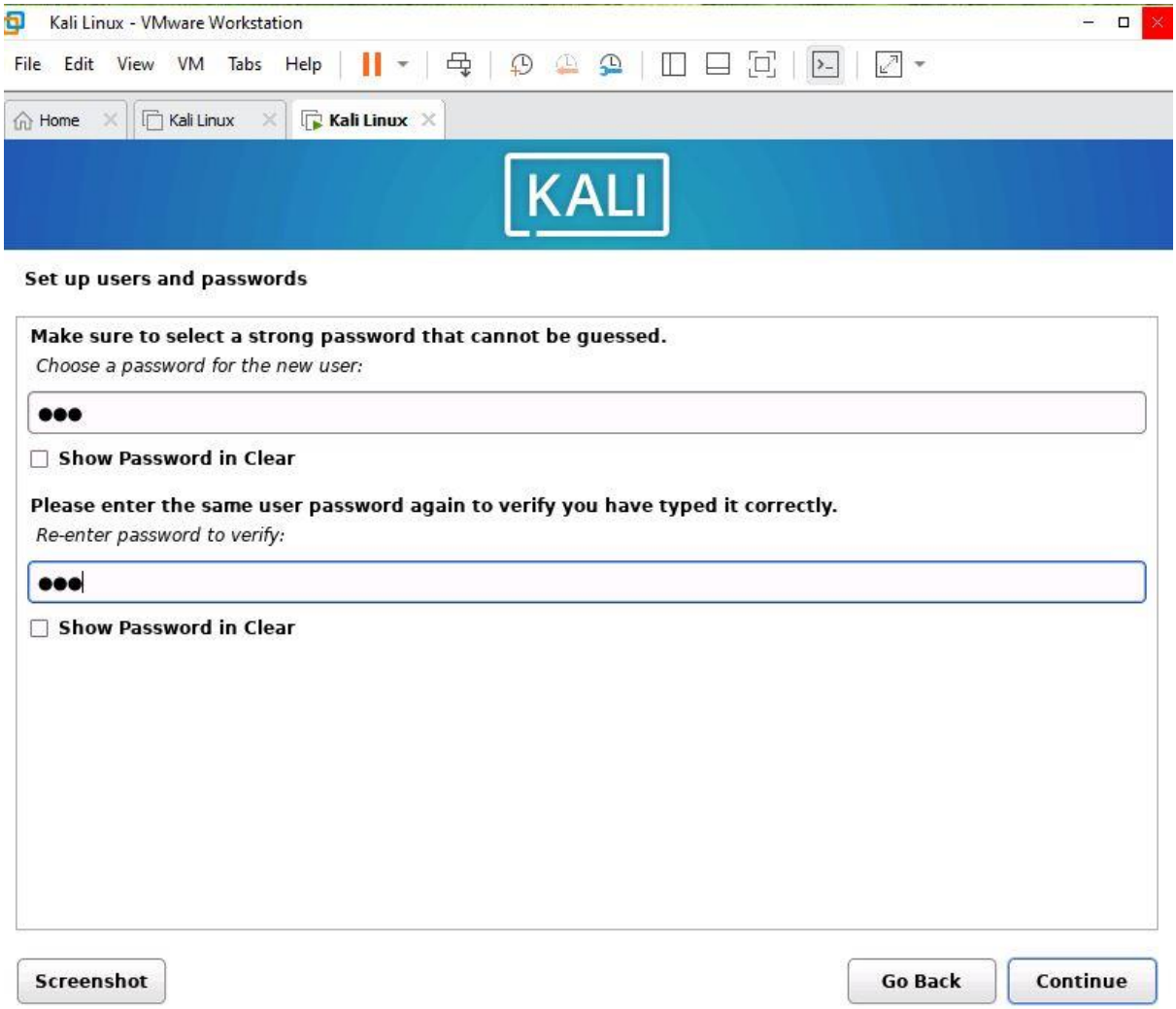



## Step 9: Set Up User Account and Password

- Root Password: Enter a strong password for the root account and confirm it. Click Continue.
- Full Name: Enter your full name for the new user account.
- Username: Enter a username (e.g., student) and click Continue.
- User Password: Set and confirm a password for this user account. Click Continue.

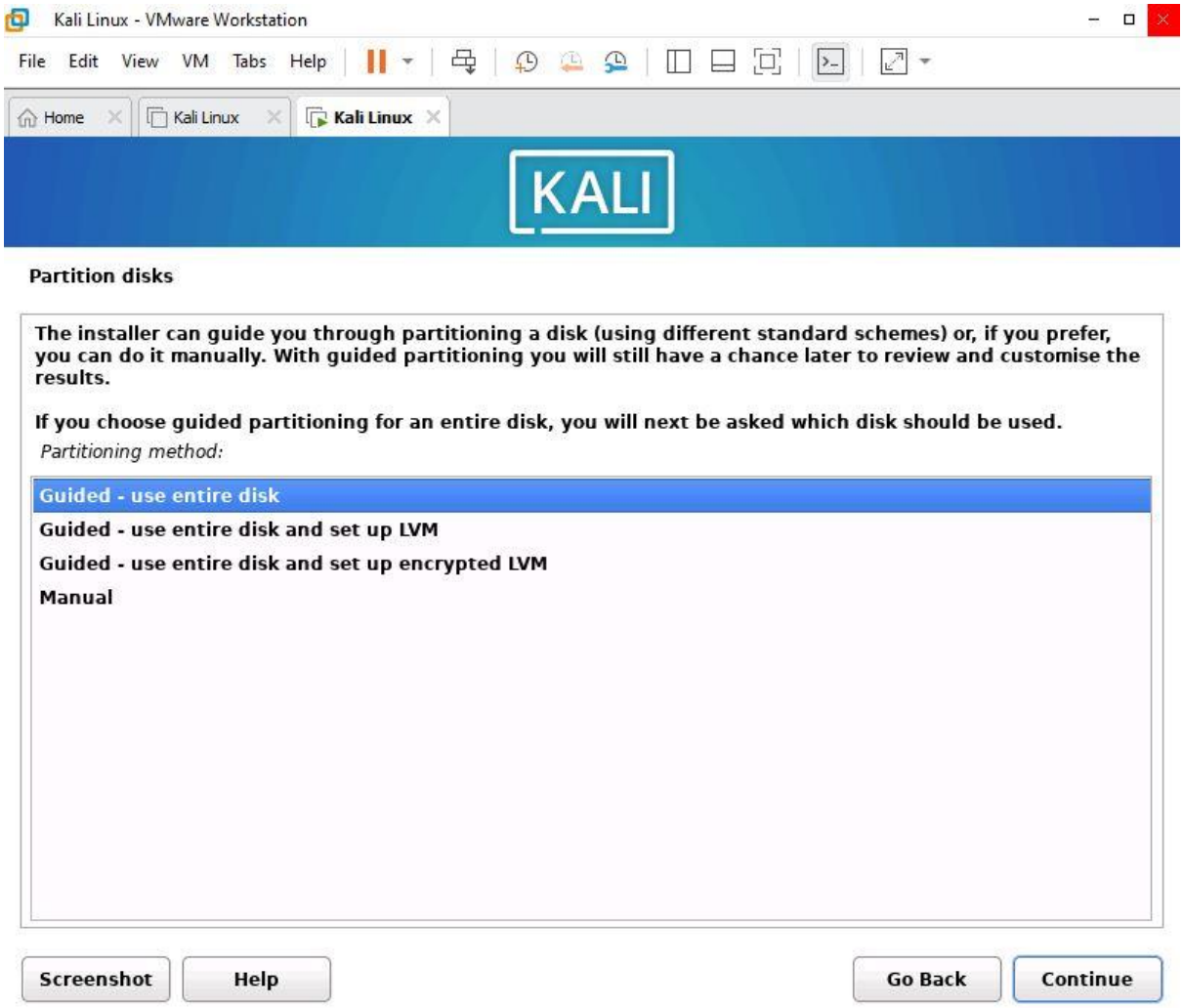
**Important:** Use a strong password containing uppercase, lowercase, numbers, and symbols. You will need this to log in.

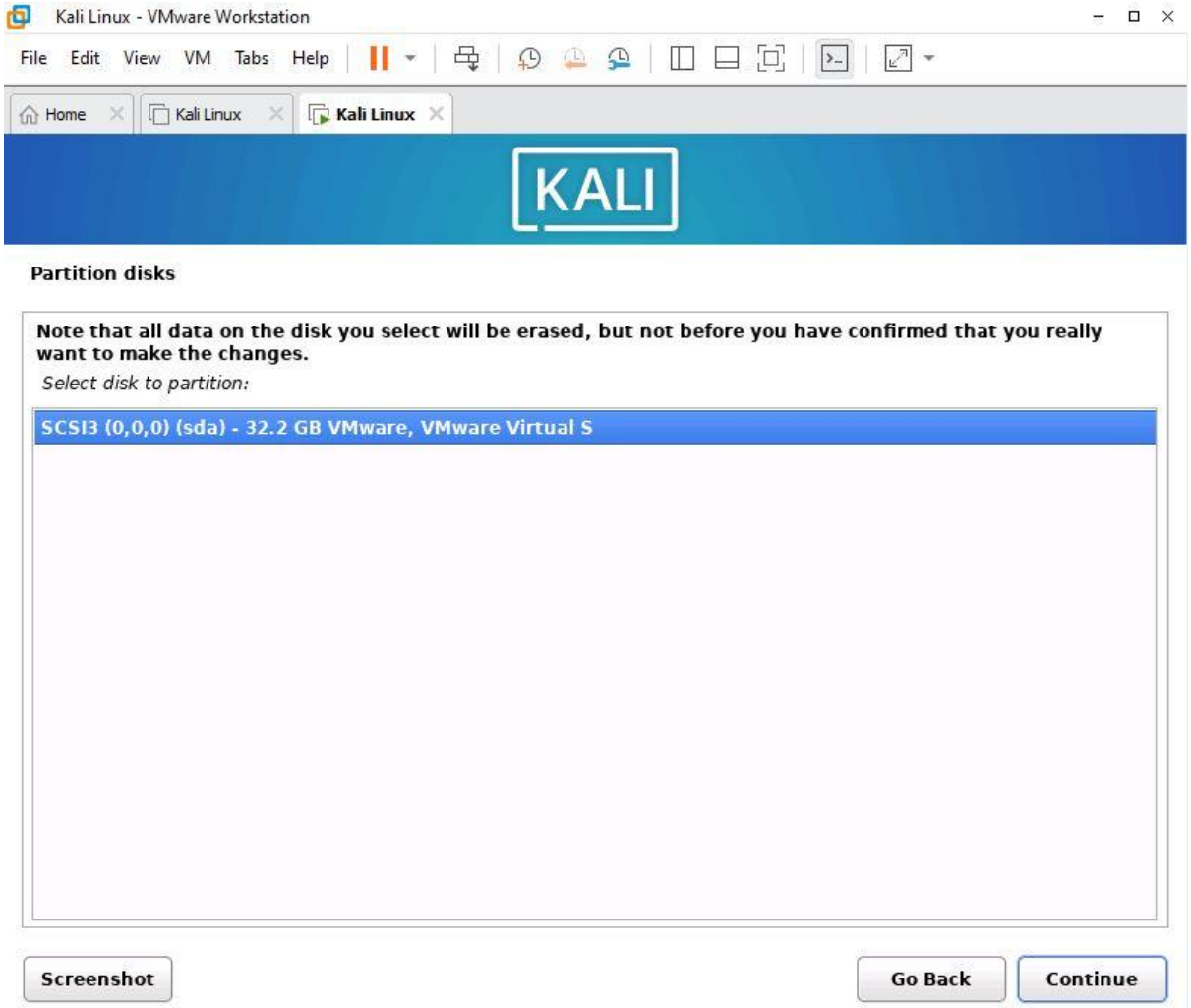

Kali Linux - VMware Workstation
File Edit View VM Tabs Help
Home
Kali Linux
Kali Linux
KALI
Set up users and passwords
Select a username for the new account. Your first name is a reasonable choice. The username should start with a lower-case letter, which can be followed by any combination of numbers and more lower-case letters.
Username for your account:
fahad
Screenshot
Go Back
Continue




## Step 10: Partition the Disk

- Partitioning Method: Select 'Guided – use entire disk' for the simplest setup.
- Select Disk: Choose the virtual disk you created (e.g., VMware Virtual S, ~30 GB).
- Partitioning Scheme: Select 'All files in one partition (recommended for new users)'.
- Review the proposed partition layout and click 'Finish partitioning and write changes to disk'.
- Confirm by selecting 'Yes' when asked to write changes, then click Continue.


Kali Linux - VMware Workstation
File Edit View VM Tabs Help
Home
Kali Linux
Kali Linux
KALI
Partition disks
The installer can guide you through partitioning a disk (using different standard schemes) or, if you prefer, you can do it manually. With guided partitioning you will still have a chance later to review and customise the results.
If you choose guided partitioning for an entire disk, you will next be asked which disk should be used.
Partitioning method:
Guided - use entire disk
Guided - use entire disk and set up LVM
Guided - use entire disk and set up encrypted LVM
Manual
Screenshot
Help
Go Back
Continue


Kali Linux - VMware Workstation
File Edit View VM Tabs Help
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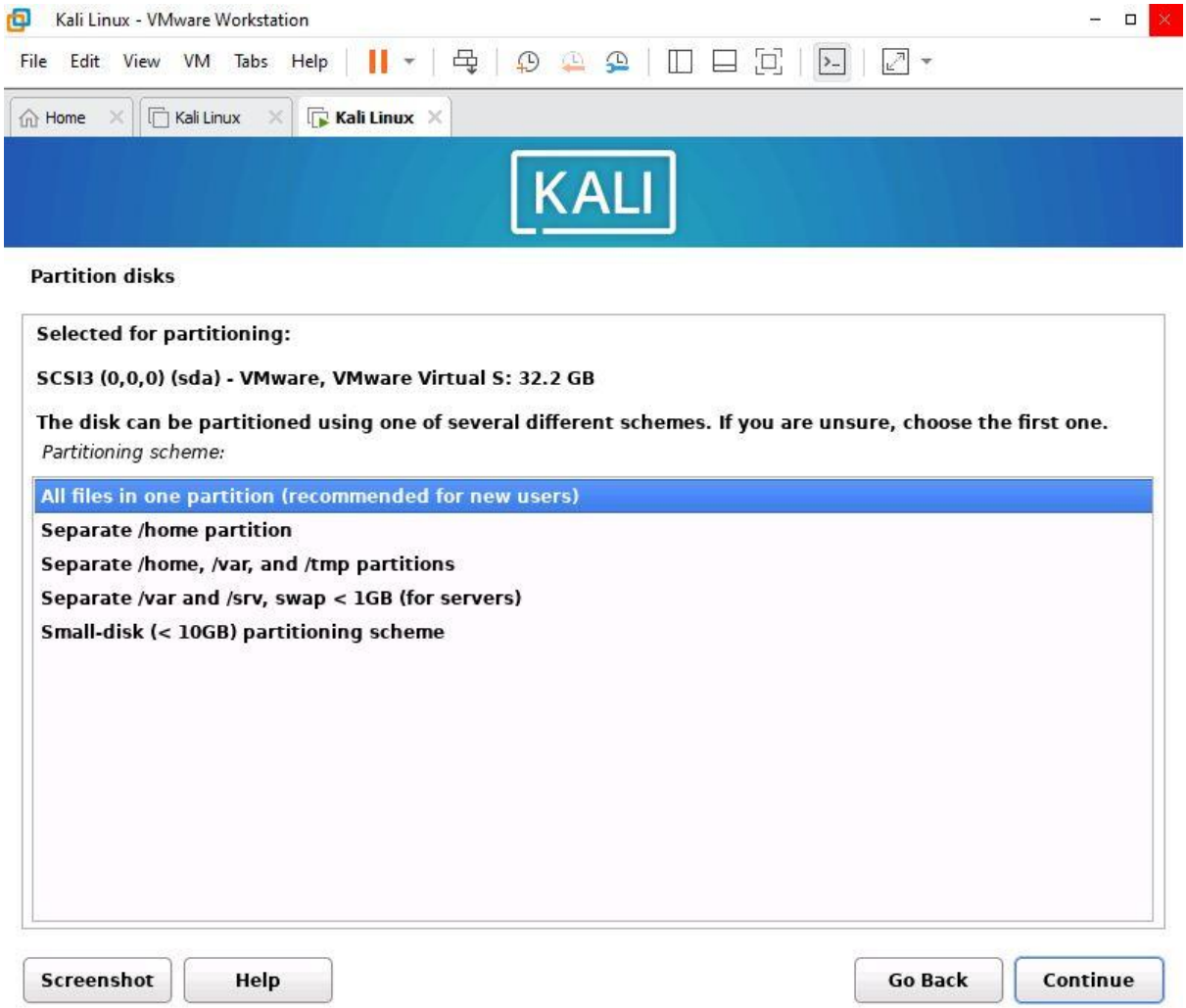
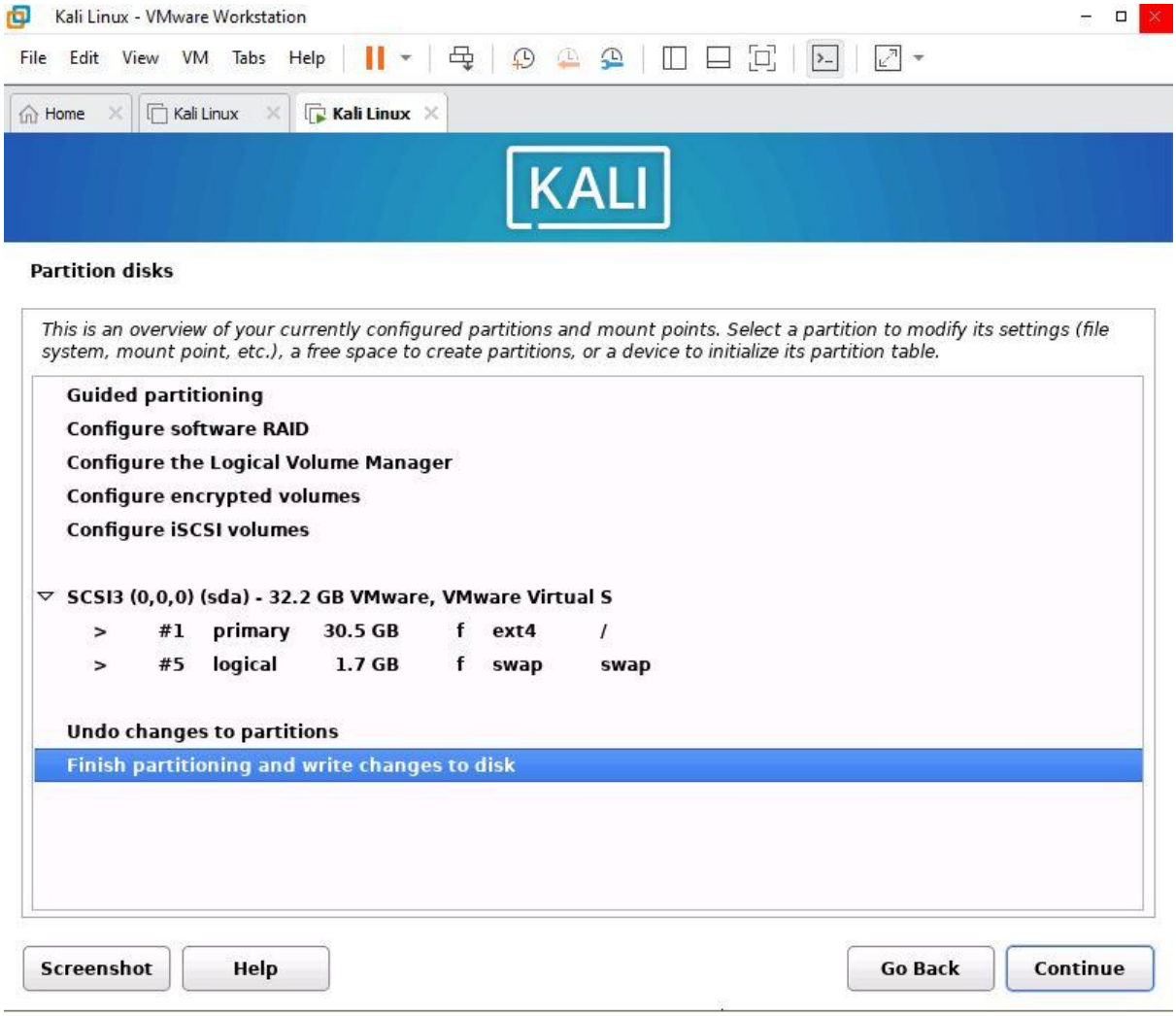
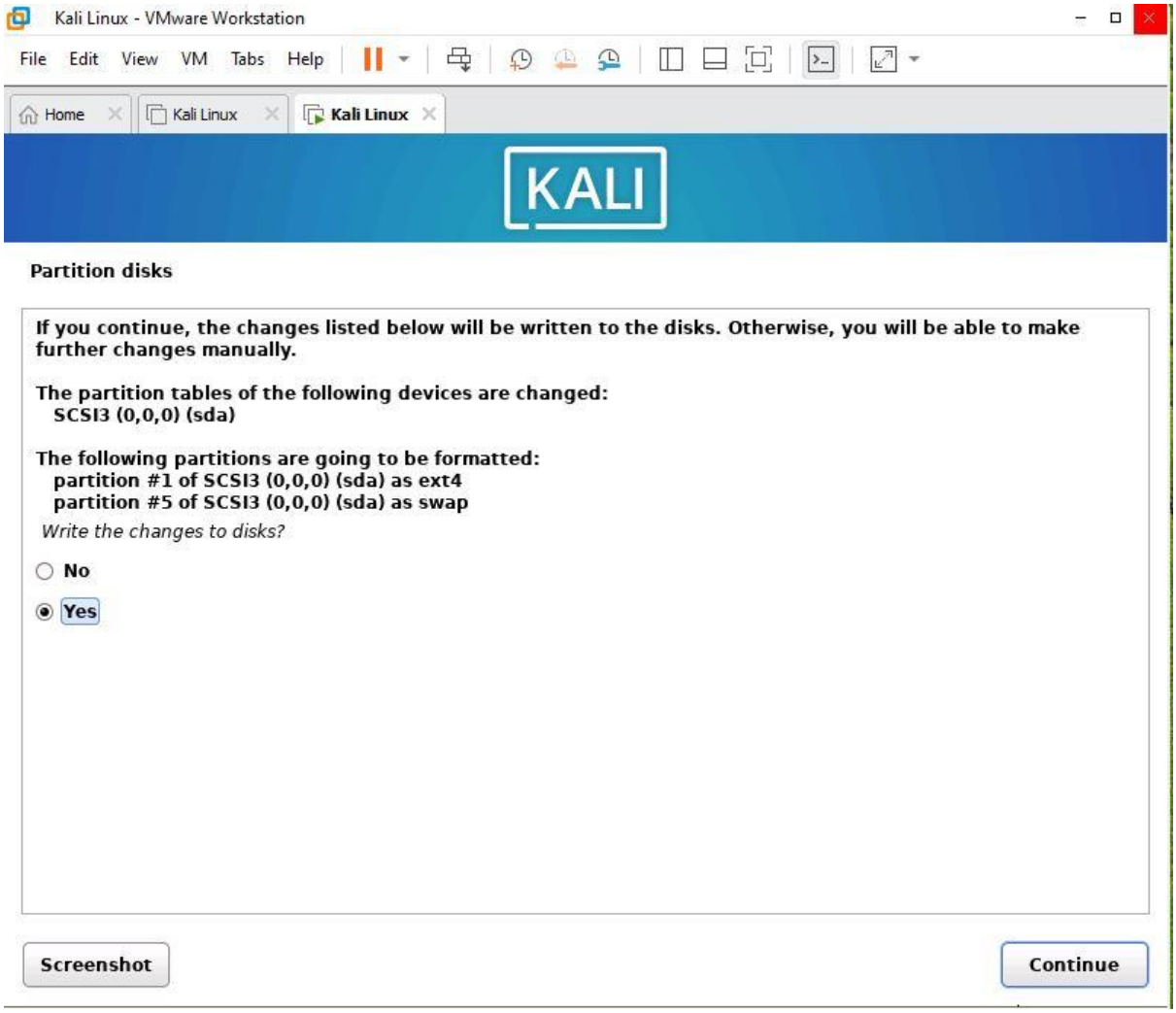
Kali Linux
Kali Linux
KALI
Partition disks
Note that all data on the disk you select will be erased, but not before you have confirmed that you really want to make the changes.
Select disk to partition:
SCSI3 (0,0,0) (sda) - 32.2 GB VMware, VMware Virtual S
Screenshot
Go Back
Continue

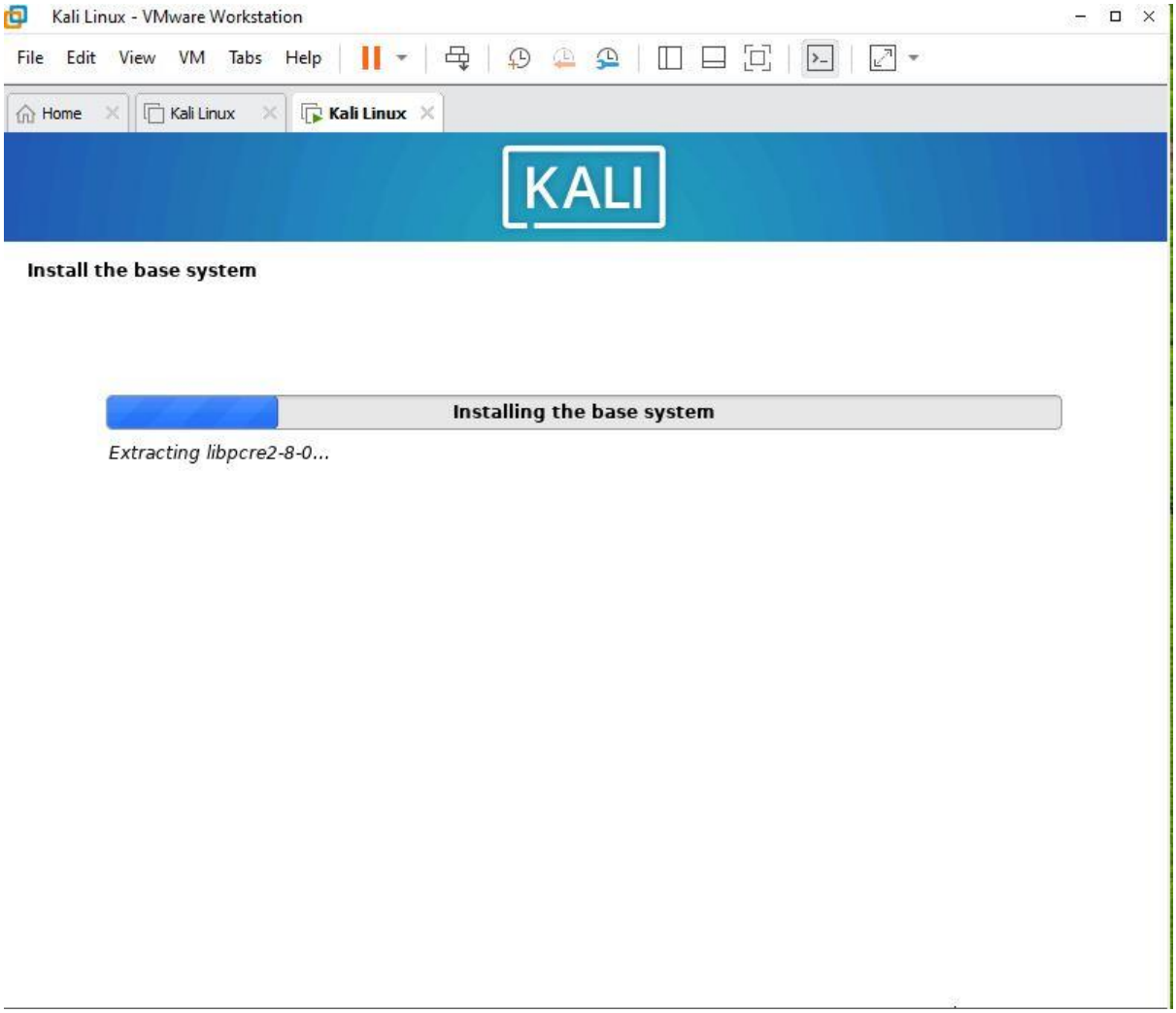

Kali Linux - VMware Workstation
File Edit View VM Tabs Help
Home
Kali Linux
Kali Linux
KALI
Partition disks
Selected for partitioning:
SCSI3 (0,0,0) (sda) - VMware, VMware Virtual S: 32.2 GB
The disk can be partitioned using one of several different schemes. If you are unsure, choose the first one.
Partitioning scheme:
All files in one partition (recommended for new users)
Separate /home partition
Separate /home, /var, and /tmp partitions
Separate /var and /srv, swap < 1GB (for servers)
Small-disk (< 10GB) partitioning scheme
Screenshot
Help
Go Back
Continue


Kali Linux - VMware Workstation
File Edit View VM Tabs Help
Home
Kali Linux
Kali Linux
KALI
Partition disks
This is an overview of your currently configured partitions and mount points. Select a partition to modify its settings (file system, mount point, etc.), a free space to create partitions, or a device to initialize its partition table.
Guided partitioning
Configure software RAID
Configure the Logical Volume Manager
Configure encrypted volumes
Configure iSCSI volumes
SCSI3 (0,0,0) (sda) - 32.2 GB VMware, VMware Virtual S
> #1 primary 30.5 GB f ext4 /
> #5 logical 1.7 GB f swap swap
Undo changes to partitions
Finish partitioning and write changes to disk
Screenshot
Help
Go Back
Continue




## Step 11: Install the Base System

- The installer will now copy and install files. This process may take 10–20 minutes depending on your hardware.
- A progress bar will show the installation status. Do not close VMware during this step.

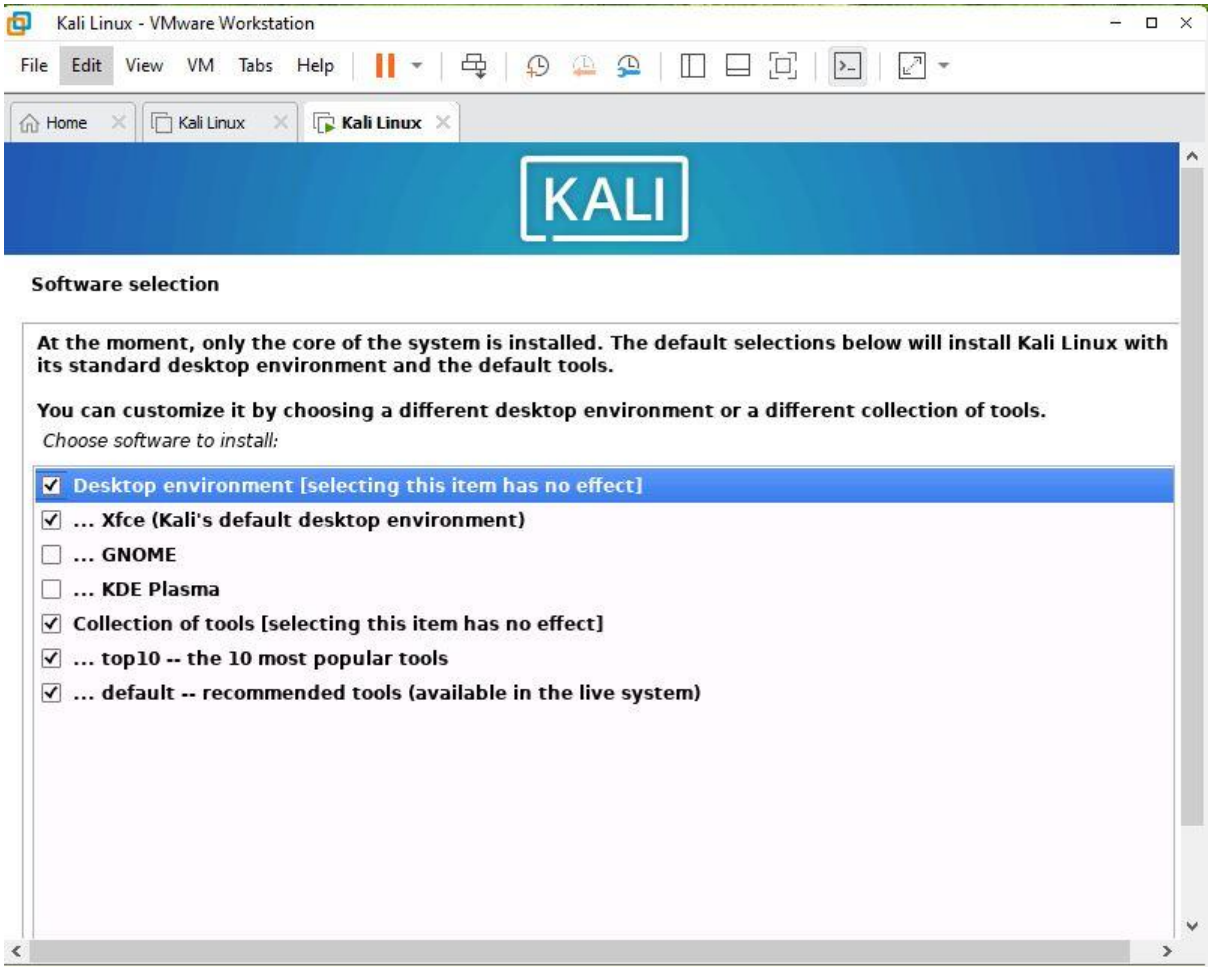



## Step 12: Configure the Package Manager (Software Mirror)

- Network Mirror: Select 'Yes' to use an online mirror for additional packages.
- HTTP Proxy: Leave blank if you have a direct internet connection, then click Continue.
- The installer will configure the package manager automatically.

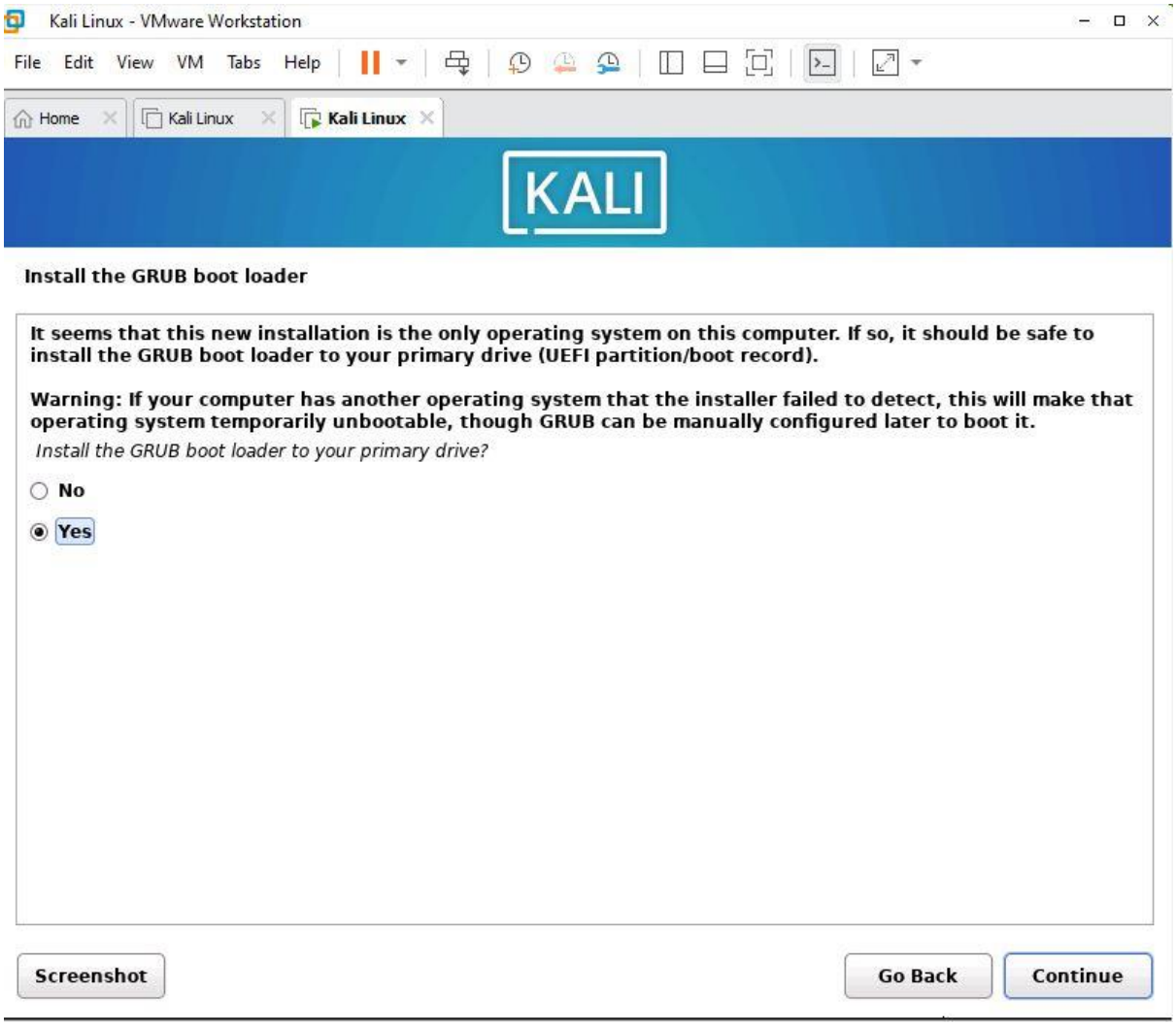
## Step 13: Select Software (Desktop Environment)

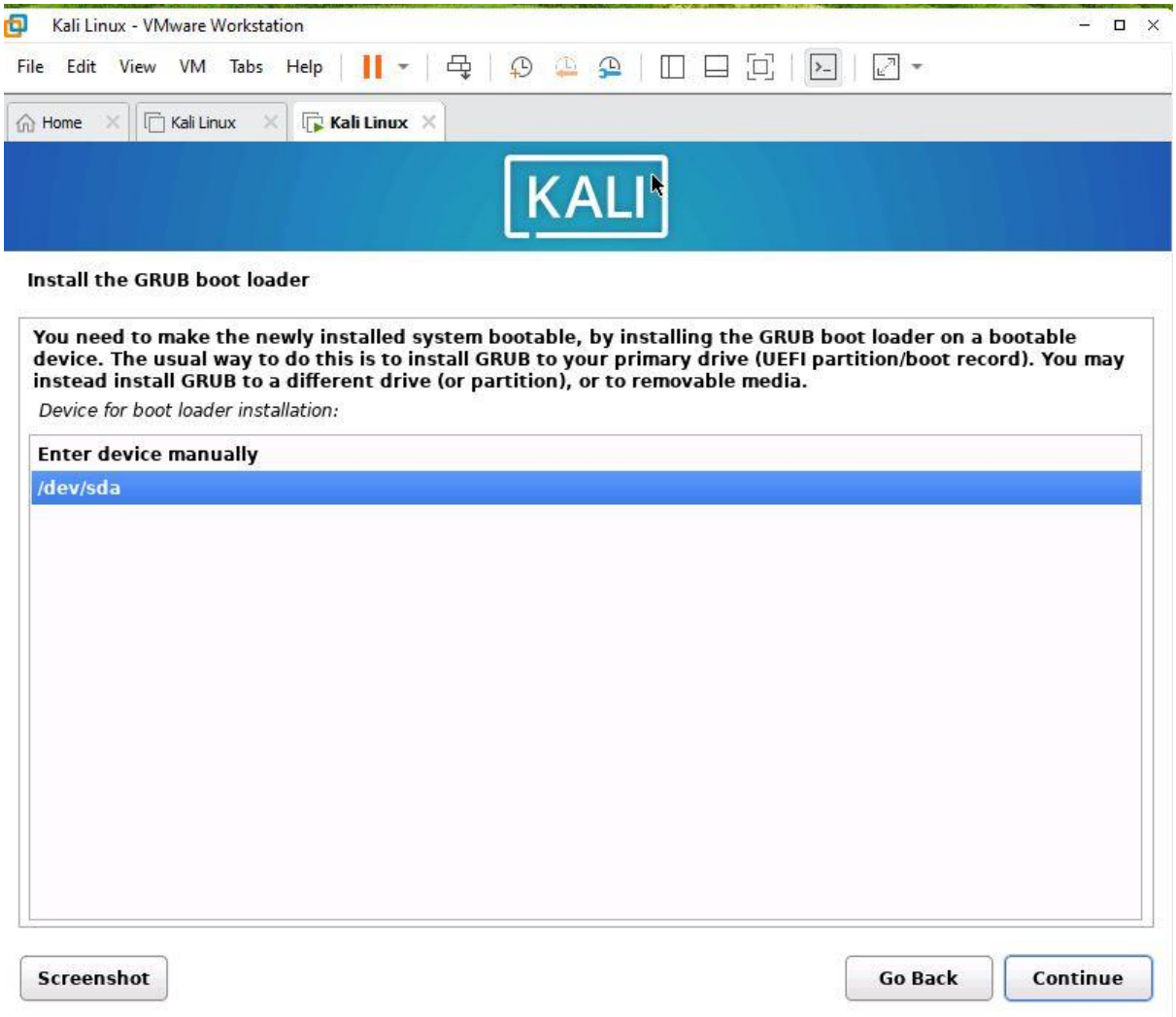
- Software Selection: A list of software options will appear.
- By default, 'Kali desktop environment' and 'GNOME' (or XFCE) are selected.
- Also ensure 'top 10' tools or 'default' tools are checked for a standard Kali setup.
- Click Continue to begin software installation (this may take 15–30 minutes).




## Step 14: Install the GRUB Bootloader

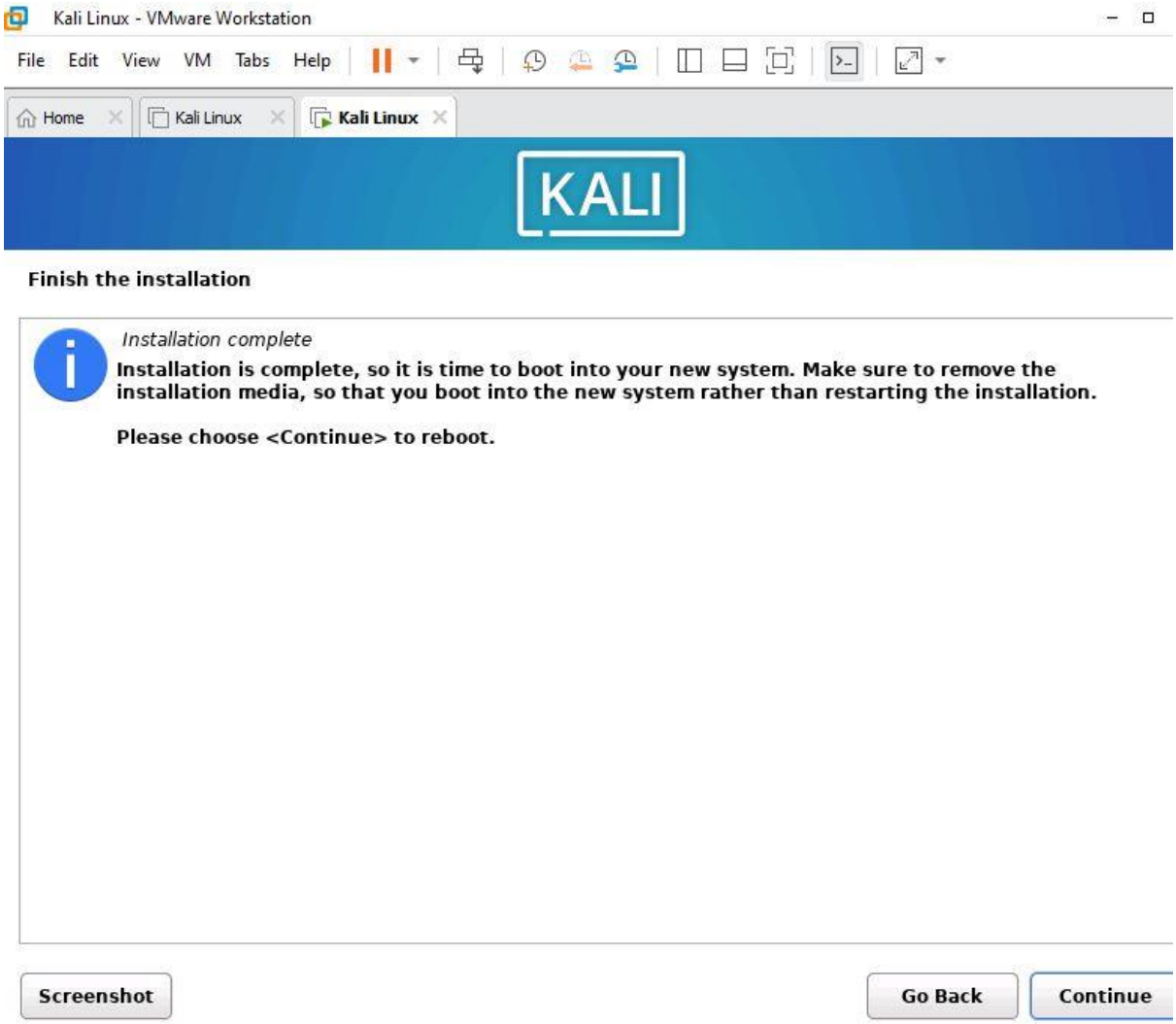
- Install GRUB: Select 'Yes' to install the GRUB bootloader to the master boot record.
- Device: Select the virtual hard disk (e.g., /dev/sda) and click Continue.
- GRUB allows the VM to boot Kali Linux after the ISO is ejected.


Kali Linux - VMware Workstation
File Edit View VM Tabs Help
Home
Kali Linux
Kali Linux
KALI
Install the GRUB boot loader
It seems that this new installation is the only operating system on this computer. If so, it should be safe to install the GRUB boot loader to your primary drive (UEFI partition/boot record).
Warning: If your computer has another operating system that the installer failed to detect, this will make that operating system temporarily unbootable, though GRUB can be manually configured later to boot it.
Install the GRUB boot loader to your primary drive?
No
Yes
Screenshot
Go Back
Continue



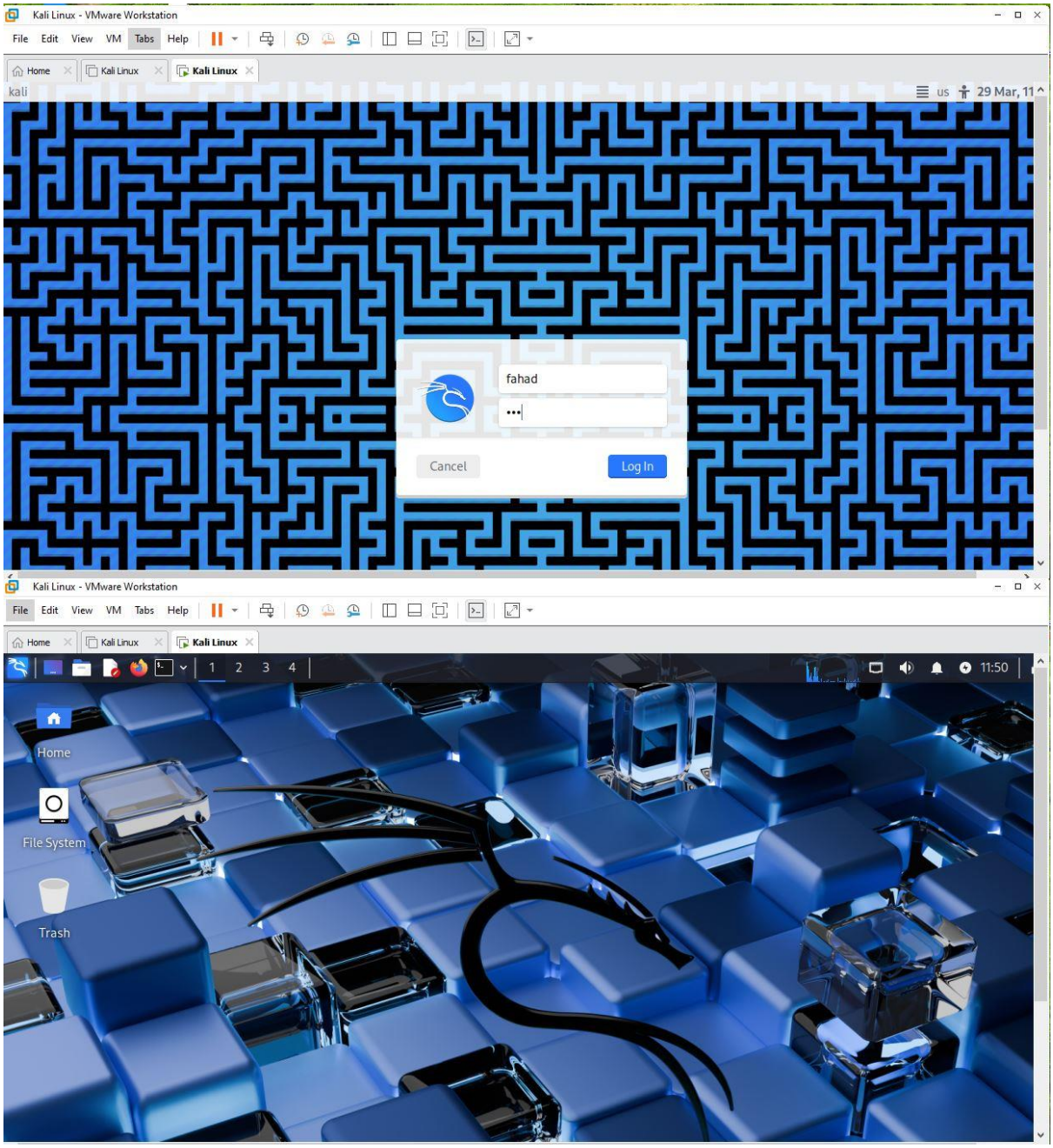

## Step 15: Complete Installation and Reboot

- The installer will display an 'Installation Complete' message.
- Click Continue to reboot the virtual machine.
- VMware will automatically eject the ISO and restart from the installed system.
- The GRUB bootloader will appear briefly, then Kali Linux will begin loading.




## Step 16: Log In to Kali Linux

- The Kali Linux login screen (GDM/LightDM) will appear.
- Enter the username and password you created during installation.
- Click 'Log In' or press Enter.
- The Kali Linux desktop environment (GNOME or XFCE) will load successfully.




## Step 17: Install VMware Tools (Optional but Recommended)

- In VMware menu, click 'VM' > 'Install VMware Tools'.
- Open a Terminal in Kali Linux (right-click desktop > Open Terminal).
- Run the following commands:

```
sudo apt update && sudo apt install -y open-vm-tools open-vm-tools-desktop
```

- After installation, reboot the VM. VMware Tools enables shared clipboard, drag-and-drop, and auto screen resizing.

```
fahad@kali: ~
Session  Actions  Edit  View  Help
┌──(fahad㉿kali)-[~]
└─$ sudo apt update && sudo apt install -y open-vm-tools open-vm-tools-deskto
p
[sudo] password for fahad:
Hit:1 http://http.kali.org/kali kali-rolling InRelease
1811 packages can be upgraded. Run 'apt list --upgradable' to see them.
Upgrading:
  open-vm-tools  open-vm-tools-desktop

Summary:
  Upgrading: 2, Installing: 0, Removing: 0, Not Upgrading: 1809
  Download size: 844 kB
  Space needed: 0 B / 12.7 GB available

Get:1 http://kali.download/kali kali-rolling/main amd64 open-vm-tools-desktop
 amd64 2:13.0.10-1 [159 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 open-vm-tools amd64 2
:13.0.10-1 [685 kB]
Fetched 844 kB in 1s (866 kB/s)
(Reading database ... 422186 files and directories currently installed.)
Preparing to unpack .../open-vm-tools-desktop_2%3a13.0.10-1_amd64.deb ...
Unpacking open-vm-tools-desktop (2:13.0.10-1) over (2:13.0.5-1) ...
Preparing to unpack .../open-vm-tools_2%3a13.0.10-1_amd64.deb ...
Unpacking open-vm-tools (2:13.0.10-1) over (2:13.0.5-1) ...
Setting up open-vm-tools (2:13.0.10-1) ...
vgauth.service is a disabled or a static unit not running, not starting it.
Setting up open-vm-tools-desktop (2:13.0.10-1) ...
Processing triggers for kali-menu (2025.4.3) ...
Processing triggers for libc-bin (2.41-12) ...
Processing triggers for man-db (2.13.1-1) ...

┌──(fahad㉿kali)-[~]
└─$
```

---

# Result

Kali Linux has been successfully installed as a virtual machine inside VMware Workstation. The VM boots into the Kali Linux desktop environment with all default penetration testing tools pre-installed. The student can now use Kali Linux in an isolated virtual environment for ethical hacking, cybersecurity labs, and practice — without affecting the host operating system.

> ✔ **Success:** The installation is complete when you can log into the Kali Linux desktop and open a terminal successfully.

## Common Issues & Troubleshooting

- VM won't boot from ISO: Verify the ISO path in VM Settings > CD/DVD > Use ISO Image File.
- Black screen after boot: Enable 3D acceleration in VM Settings > Display or disable it if already on.

- Slow performance: Increase RAM or CPU cores in VM Settings, and install VMware Tools.
- No internet in VM: Ensure network adapter is set to NAT in VM Settings > Network Adapter.
- GRUB not found after install: Boot from ISO again and re-install GRUB to /dev/sda.